A FIELD REPORT ON
LEVELLING, TRAVERSING, TACHEOMETRY, TRIANGULATION, INTERSECTION, RESECTION, CHAIN AND COMPASS SURVEYING

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ABSTRACT

This is a field report regarding the general survey procedures. All the surveying techniques like Levelling, Traversing, Tacheometry, Triangulation, Intersection, Resection, Chain and Compass surveying detailing and plotting those coordinated on tracing paper and permatrace are done during this Project. All the surveying works are done systematically following survey procedure.

Finally, the computed data were adjusted and used for detailing as well as plotting the map and graph. Under this project we have established control points, drawn offsets, determined elevation change and performed resection as well as intersection.

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LIST OF ABBREVIATIONS

- BS = Back Sight
- BM= Bench Mark
- D-CARD= Description Card
- E= Easting
- FS = Foresight
- HI= Height of Instrument
- IS = Intermediate Sight
- MSL= Mean Sea Level
- N= Northing
- PBM = Permanent Benchmark
- RL = Reduced Level
- TBM = Temporary Benchmark

CHAPTER ONE: LEVELLING

1.1 Background:

Levelling is defined as the art of determining the relative heights or the elevations of different points on the surface of the earth so that the same may be represented on a plan or a map. Levelling process mainly deals with the measurement in the vertical plane.

1.1.1 Principle of levelling:

The principle of leveling in surveying involves measuring and establishing elevations of points on the Earth's surface relative to a known benchmark using a leveling instrument. It relies on the concept of a horizontal plane and precise height differences between points to determine accurate elevations.

1.1.2 Types of levelling:

a. Direct levelling:

i. Simple levelling

Simple levelling is the simplest operation in leveling. It is done when it is required to find the difference in elevation between two points. The both position of staff must be visible single position of level. The reading can be obtained on a staff held successfully upon these points.

ii. Differential levelling

Determining height difference between two points which are far apart horizontally and vertically, the height difference between them cannot be determined by one instrument setup.

iii. Reciprocal levelling

Reciprocal leveling is a surveying technique used to enhance the accuracy of elevation measurements between two points. It involves taking leveling readings in both directions (from A to B and then from B back to A) using a leveling instrument. By averaging the readings from both directions, errors due to instrument or observation inaccuracies can be minimized, resulting in more precise elevation data.

iv. Fly levelling

It is a type of differential levelling in surveying done to determine approximate elevations of different points. The fly levelling is done where rapidity, but low precision is required. Fly levelling is generally used for the reconnaissance of the area or for approximate checking of the levels.

v. Profile levelling

Profile leveling is a surveying technique used to determine the elevations of points along a line. It involves measuring vertical distances between points and calculating changes in elevation to create a profile of the terrain or structure being surveyed. This method is essential for designing roads, railways, pipelines, and other infrastructure projects where accurate elevation data is critical.

vi. Precise levelling

Precise leveling is an advanced surveying method used to accurately determine elevations over longer distances with high precision. It involves meticulous measurements using precise instruments like digital levels or total stations, ensuring minimal error in elevation calculations.

vii. Check levelling

Check leveling is a verification process used in surveying to ensure the accuracy of previously conducted leveling measurements. It involves re-measuring selected points along a survey line using the same or different instruments to compare with initial measurements. This helps identify any discrepancies or errors in the original leveling data, ensuring the reliability and precision of the elevation information gathered for engineering and construction projects.

b. Trigonometric levelling

Trigonometric leveling is a surveying technique that determines elevations using trigonometric principles rather than traditional leveling methods. It involves measuring horizontal and vertical angles to calculate elevations indirectly. This method is particularly useful in areas where direct leveling is impractical, such as rough terrain or across bodies of water. Trigonometric leveling can provide accurate elevation data over longer distances and challenging landscapes, making it valuable for engineering and mapping applications.

c. Barometric levelling

Barometric leveling is a surveying technique that estimates elevation differences between points using atmospheric pressure measurements, which vary with altitude. It's useful for rough elevation estimates over short distances, especially in remote or challenging terrain where traditional leveling methods are impractical.

d. Stadia levelling

Stadia leveling is a surveying method that uses stadia rods equipped with stadia hairs or markings to measure vertical distances. By observing the apparent spacing of these markings through a surveying instrument, engineers can calculate elevations accurately. This method is valuable for quick and precise elevation assessments in various terrains, such as construction sites or rugged landscapes.

1.1.3 OBJECTIVE

The primary objective is:

- To establish a network of elevation benchmarks across [specific area or project site].

The secondary objectives are:

- Instrument handling.
- To prepare graph of profile and cross section levelling.

1.1.4 Study Area

Starting from the known elevation at PBM 1 (near the bridge), we carried our project along the line of progress to the PBM 2 (near chiyan dada).

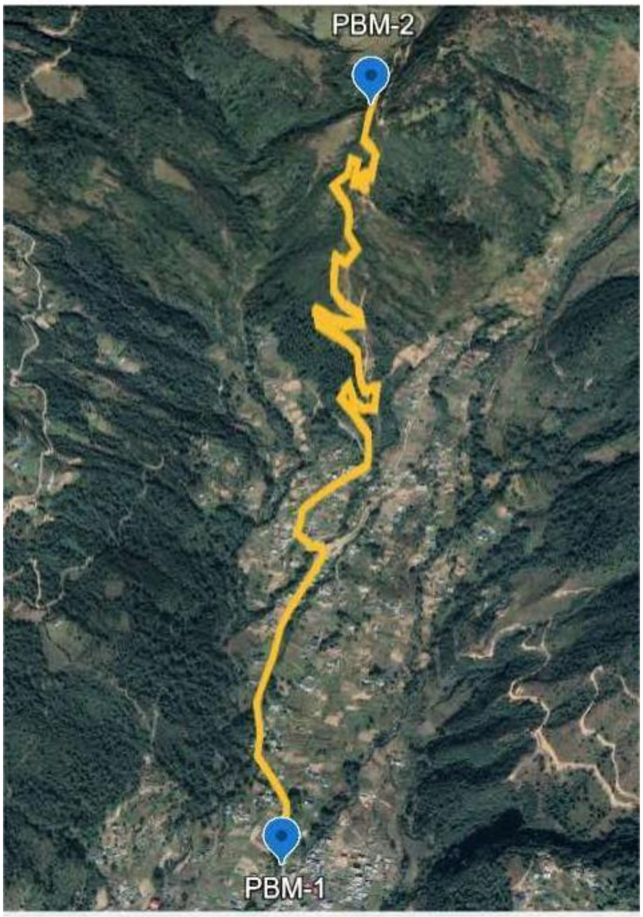

Figure 1: Study area of levelling

1.1.5 SCOPE OF THE WORK

The objective of our project was to establish vertical control points and calculate the elevation difference. It can be used to find the gradient of the road which is useful for canal and drainages. Since it was a fourth order levelling, permanent monumentation was not done. So, the benchmarks aren't useful in future.

1.2 Literature Review

1. **Datum:**

   Datum is an imaginary level surface or a level line from which the vertical distance of different points are measured.

2. **Mean Sea Level:**

   The average height of the sea's surface for all stages of the tide over a very long period (usually 19 years).

3. **Elevation:**

   Elevation is used to describe the height of a place above the sea level.

4. **Altitude:**

   Altitude is used to describe the vertical distance between an object and a reference point (Earth).

5. **Reduced Level:**

   The reduced level of a point is its height relative to the datum. It is the calculated height of the point above or below the datum. The reduced level is used synonymously with the term elevation.

6. **Level surface:**

   Any surface parallel to the mean spheroidal surface or the curved surface of the earth is known as level surface. It is the curved surface which at each point is perpendicular to the direction of gravity.

7. **Level line:**

   Any line lying on a level surface is level line. This line is normal to the direction of gravity and the plumb line at all points.

8. **Horizontal plane:**

   Any plane tangential to the level surface at any point is known as horizontal plane.

9. **Horizontal line:**

   Any line lying on the horizontal plane is said to be horizontal line.

10. **Benchmark:**

    Benchmark refers to the point of known elevation above or below datum. They serve as reference points for finding the RL of new points. There are two types of benchmarks.

- Permanent Benchmark (PBM): These are the permanently established points of known elevation with the view that these would come for future use.
- Temporary Benchmark (TBM): The benchmark which are established temporarily for certain project and is not intended to use for other projects are known as temporary benchmark.

11. Back sight:

It is the reading taken on a staff held at a point of known elevation or at the point whose elevation has already been determined.

12. Foresight:

Foresight is the last sight reading taken from any instrument position before shifting the instrument.

13. Intermediate sight:

An intermediate sight (I.S) is any staff reading taken on the point of unknown elevation after the back sight and before the foresight. This is necessary when more than two staff readings are taken from the same position of the instrument.

14. Height of instrument:

The elevation of the line of sight with respect to the assumed datum, is known as height of instrument.

HI = BS+RL of BM

15. Turning point:

For levelling over a long distance, the instrument has to be set up a number of times. The turning point is the point selected on the route before shifting the instrument. The turning point is also called a change point (C.P).

1.3 Methodology

1.3.1 Data used

Reduced level of PBM 1 was used in this project. The R.L used was 1925m.

1.3.2 Instrument used

- **Dumpy level:** Dumpy level is the most also called as automatic level or builder's level. Elevations of different points and distance between the points of same elevation can be determined by dumpy level. The telescope is fixed to its supports in dumpy level and hence it cannot be rotated in vertical axis.
- **Levelling staff:** A leveling staff is a measuring tool used in surveying to determine vertical distances or elevations, marked with precise measurements for accurate height reading.
- **Meter tape (synthetic tape):** A meter tape is a flexible tool used for measuring distances, marked with centimeter and meter graduations. It's commonly used in surveying, construction, and other applications requiring precise length measurements.

1.3.3 Method

- Planning

  Firstly, suitable loop was chosen around jiri to carry our levelling project and our initial and final point was determined. Vertical height of one PBM was collected.

- Reconnaissance and Monumentation

  On the first day of levelling, the project site was visited and studied the feasibility of the area for levelling. 2 permanent benchmarks and 10 temporary benchmarks were established considering the length. The benchmarks were monumented by using wooden peg with cross mark on its top.

- Observation

  The total route was divided into 11 loops. The top, middle and bottom reading of staff was observed on both backsight and foresight which was later used to determine the elevation change and horizontal distances.

- Computation

  For elevation change, Rise and Fall method was used. In this method, the foresight reading is subtracted from backsight reading.

  If Backsight – Foresight > 0, then it is considered as Rise.

  If Backsight – Foresight < 0, then it is considered as Fall.

  The Rise is added on R.L and Fall is subtracted from R.L.

  After the computation of R.L, arithmetic check was done.

  $\sum$B.S - $\sum$F.S = $\sum$Rise - $\sum$Fall = Last R.L – First R.L

- Error Distribution and Adjustment

  The error was distributed with respect to the distance between two stations. If the error is positive then the correction to be applied is negative. If the error is negative then the correction to be applied is positive.

  Correction in each station = ± (closing error in the loop * distance between each setup of instrument) / total distance of each loop.

- Plotting the profile

  After the computation and the error adjustment, L-section graph was plotted with scale as Horizontal scale = 1:2000

  Vertical scale = 1:200

  Also, cross section graph was plotted with scale as

  Horizontal scale = 1:2000

  Vertical scale = 1:200

1.4 Result and Discussions

From the project, following results were obtained:

- Elevation changes between the benchmarks.
- Reduced level of PBM and TBM.

Table 1: Data for the PBMs and TBMs R.L calculated.

STATION	REDUCED LEVEL(m)	DISTANCE(m)
PBM1	1925	0
TBM1	1952.610	257.3
TBM2	2002.566	420.9

TBM3	2060.412	503.1
TBM4	2099.675	309.7
TBM5	2127.575	240.9
TBM6	2156.134	192
TBM7	2184.641	254.5
TBM8	2214.509	212.9
TBM9	2238.849	219
TBM10	2261.124	170.6
PBM2	2272.798	94.6

Below are the L-section graph obtained from the calculated data from the project.

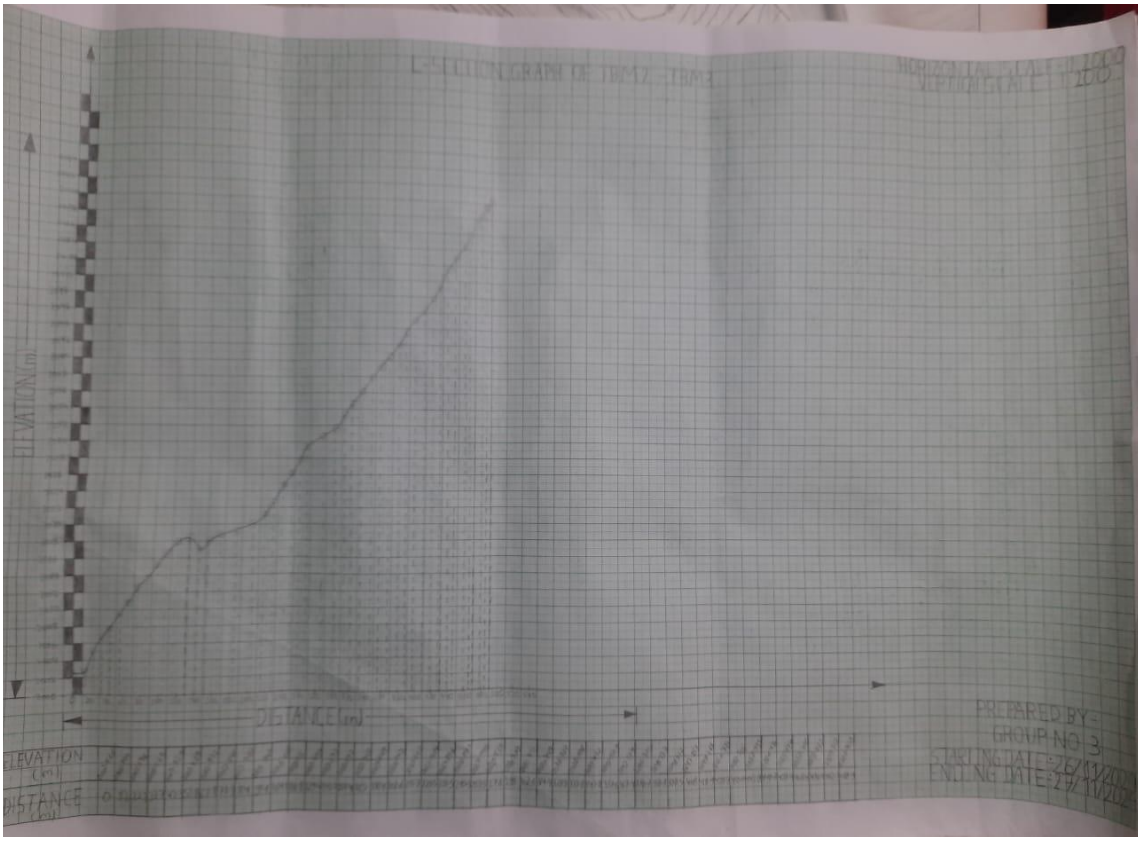

Figure 2: L-Section graph from TBM2 to TBM3

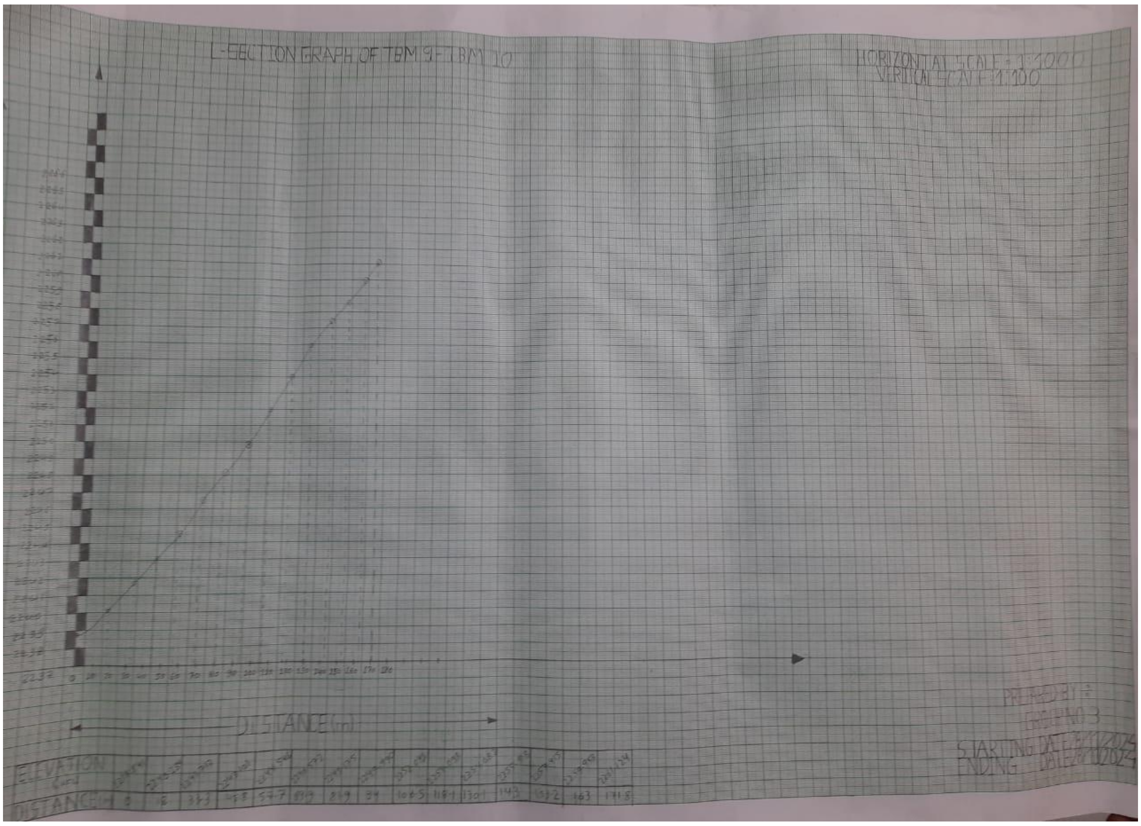

Figure 3: L-Section Graph from TBM9 to TBM10

1.5 Conclusions and Recommendations

We calculated various data for depression and elevation. These data were used to compute reduced level of PBMs and TBMs and were also used in L-section graph. Difference in elevation between two PBMs was found to be 347.798m.

This project helped us to work in team and developed the leading ability. Various instruments were used and handled. A real working experience was collected. To conclude, base for future extension of the road was carried out with personal experiences.

1.6 References

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2 CHAPTER TWO: TRAVERSE

2.1 Introduction

2.1.1 Background

Traversing is the type of survey in which the length and direction of a series of connected lines are measured with the help of distance and angle measuring instruments respectively. Generally, the linear measurements are made with the help of tape, chain or EDM, and the relative direction of the lines are measured with a compass or theodolite. It is a technique of establishing horizontal control points in dense and narrow areas where second order triangulation is not possible.

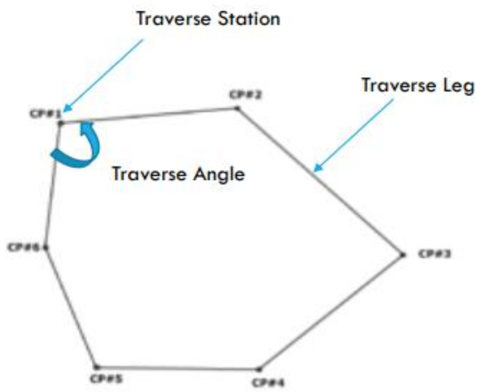


Figure 4: Traverse loop

Traversing has a wide application in various fields of geospatial survey procedure. Some of the main purposes of traversing incorporate determining existing boundary lines, calculation of area within the boundary, establishing control points for photogrammetric surveying and mapping, establishing control points for calculating earthwork quantities and so on.

Principle of traverse surveying:

The principle of traversing is that if the direction and length of any line is known then the coordinates of its head (end point) can be found from the given coordinates of its tail (starting point).

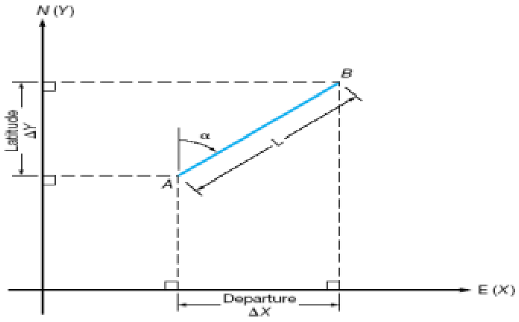


Figure 5: Departure and Latitude of a line

Latitude (ΔN) = d cosα

Departure (ΔE) = d sinα

If the coordinate of A is known.

Then the coordinate of B can be calculated.

i.e. $E2 = E1 + \Delta E$ and $N2 = N1 + \Delta N$

2.1.2 Types of traverse surveying

Traverse surveying is classified on the basis of following headings:

A. On the basis of starting and ending point

B. On the basis of instrument used

C. On the basis of accuracy and precision

A. Traverse based on starting and ending point:

Further on the basis of starting point and ending point, traverse can be divided into 2 divisions:

a. Closed loop traverse: It starts either from a known starting point and ends at the same station or another known station. It has more checks and it is thus more accurate. It is itself a linear and angular check. It is suitable for the survey of boundaries of ponds, sports grounds, forests, etc.

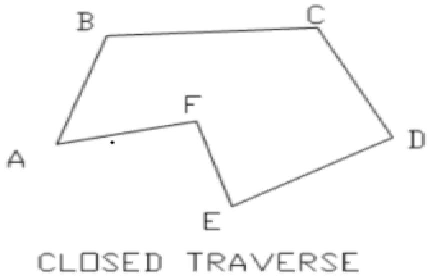


Figure 6: Closed loop traverse

b. Open loop traverse:

It either starts from a known or unknown station and ends at another unknown station or known station. It is geometrically and mathematically open. It checks only possible by additional observation. It has no check itself and is used in linear alignment. It is mostly suitable for long narrow strips. For example: it is used for surveying of roads, coastal lines.

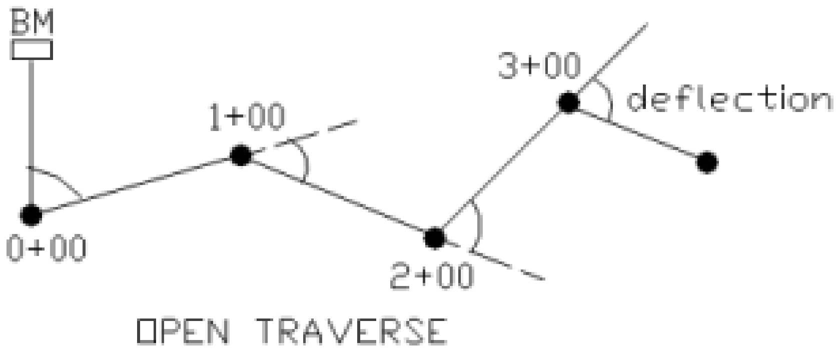


Figure 7: Open loop traverse

B. Traverse based on instrument used:

a. Chain traverse

In this traverse, only linear measurements are taken. Angles are preserved or set out by a tie line. It is mainly used for flat, level and small areas.

b. Compass traverse

Magnetic Compass is used for measuring direction of leg and tape is used for distance measurement. It is used for small, busy ground where chain traverse is not possible. Bowditch's method is used to adjust traverse.

c. Plane table traverse

In this traverse, alidade is used for direction of traverse leg and distance measured by tape. It is used for a comparatively larger area than a compass. In this traverse, adjustment is done graphically.

d. Theodolite traverse

In this traverse, theodolite is used for angular measurement and steel band is used for linear measurement. It helps in precise and accurate control point establishment. The adjustment in this traverse is done by Transit rule/Bowditch rule.

e. Tacheometry traverse

In this traverse, angle measurement is the same as in theodolite traverse and the distance is measured by stadia method.

C. Traverse based on accuracy and precision:

a. Primary traverse:

Primary traverse is highly precise traversing. Long range EDM is used for distance and precise theodolite T2/T3 is used for angle measurement in this traverse. It is used where second order triangulation is not possible. In this traverse, 5km of traverse legs are taken into account.

b. Secondary traverse:

Secondary traverse is conducted under controlled area of primary traverse. It is used to establish the third and fourth order points. In this traverse, 5km of traverse leg is taken into account.

c. Tertiary traverse:

Tertiary traverse is conducted under controlled area of primary and secondary order traverse. It is used for cadastral purposes.

2.1.3 Objectives

The main objective of traversing project was:

- To establish horizontal and vertical control points of the fourth order tacheometric surveying around Jiri area using theodolite.

The secondary objectives achieved during the accomplishment of traversing are pinpointed below:

- To provide the framework for taking details to prepare the Topographic maps.
- Learned about measuring distance, bearing and exterior angles of the traverse leg.
- Learned about setting up the monuments in the field.
- Preparation of D cards
- Measurement of horizontal and vertical angle using theodolite.

2.1.4 Scope

The course of traversing was basically focused on establishing control points. It dealt with the scope of determining cultural features of the traverse stations (i.e. (x, y, z) coordinates). We had established 11 control points by the method of traverse. The monumentation used for stations are of temporary nature (wooden pegs are used). So, permanent monumentation of the control point is not under the scope of the project. We carried out theodolite traversing. Hence, traversing by Plane Table, Chain and Compass is not under the scope of our project. Furthermore, preparation of D-Cards of monuments are done for future reference of those established control points.

2.2 Literature Review

Technical terms used in traversing are as follows:

- **Traverse leg:** The straight line between two consecutive traverse stations is called a traverse leg.

- **Bearing:** The bearing of a line is the horizontal angle which it makes with a reference line (meridian).

- **Closing error**: The distance by which a traverse loop fails to close is known as closing error or error of Closure.

- **Linear misclosure**: The discrepancies that represent the difference in ground between the positions of the point computed from the observations and the known position of the point is called Linear Misclosure.
  i.e., $eLin = \sqrt{(eLat)^2 + (eDep)^2}$
  Where,
  eLat is Northing misclosure and eDep is Easting misclosure.

- **Permissible misclosure:** The allowable amount of error in the certain observation is known as permissible misclosure. It differs according to the order of observation being carried out.

- **Traverse adjustment:** the mathematical process of eliminating the closing error by distributing them among lines by appropriate mathematical rule/principle.

- **Transit method of angular adjustment:** Transit method of angular adjustment is used where angular measurements are more accurate than linear measurement. This method states, the correction to latitude or departure of any traverse leg should be proportional to latitude or departure instead of length traverse leg. i.e.
  Correction of latitude of traverse leg= Total error in latitude(Latitude of that traverse/Total sum of latitudes*
  Similarly, Correction of departure traverse leg= total error in departure(latitude of that traverse leg total sum of departure)*
- **Bowditch method of angular adjustment:** Bowditch method of angular adjustment is used when angular and linear measurements are equally precise. According to Bowditch method,
  Adjustment of latitude in AB=Latitude misclosure ∗ (Length of AB/Perimeter of Traverse)
  Similarly, Adjustment of Departure in AB= (Departure misclosure ∗ Length of AB/Perimeter of Traverse)

Graphical method of angular adjustment: In the graphical method of adjustment, the traverse is directly plotted from field observations on sufficiently large scale after adjusting the traverse angles or the bearings of traverse legs to satisfy the geometric conditions of the traverse.

1. **Linear accuracy:** Linear accuracy is defined as the ratio of closing error to the perimeter of the traverse leg.

2. **Horizontal angle:** Horizontal angle is the angle made between two traverse legs.

3. **Zenithal angle:** It is the angle made by the traverse line with zenithal.

4. **Angular accuracy:** The numerical difference between the computed checks and the measured sum is called angular Accuracy.

5. **Traversing:** A traverse is a series of connected lines whose lengths and directions are measured in the field. The survey performed to evaluate such field measurements is called traversing.

6. **Traverse station**: The established station on connected lines is called traverse station.

7. **Meridian:** The fixed reference line about which bearing is measured is called meridian.

8. **Index error:** when the instrument in face left, the vertical circle should read 90˚or 270˚, when the line of sight is horizontal, if it does not the deviation is known as index error.

9. **Azimuth:** True bearing of a line is azimuth.

10. **Latitude:** The latitude of a line is its projection on the north-south meridian and is equal to the length of the line times the cosine of its bearing.

11. **Departure:** The departure of a line is its projection on the east-west meridian and is equal to the length of the line times the sine of its bearing. As we know that the main purpose of traversing surveys is to establish horizontal control points. So, by applying the main principle of traversing we can determine the successive control points of unknown points by using the coordinate of known points.

12. **Control Points:** Control points are the mathematical position of location (x, y, z) with respect to a reference datum used for surveying and mapping.

Applicability of Traverse

- Traverse is based on the principle of surveying i.e. working from whole to part.
- It forms the framework for tacheometric surveying.
- If the coordinates and bearing of one traverse leg is known then the other coordinates can be determined.
- Area of the closed loop can be determined.

Selection of Traverse station

For the appropriate selection of traverse station following things must be taken in consideration:

- As far as possible, we must try to work from whole to part.
- No, the station should be inter-visible.
- Stations should be selected according to requirement of work in such a way that they facilitate the survey work.
- Station should be inter-visible.
- Station should be selected on the firm ground so that instrument does not settle during taking observation.
- Station should be located in such a way that from the traverse lines all the details of the area can be plotted.

Traverse based on Accuracy and precision

Traverse is divided into three types on the basis of precision:

1 Primary Traverse: It is highly precise traversing. Total station is used for the distance and precise theodolite T2/T3 is used for angle measurement.

2 Secondary Traverse: It is under the controlled area of primary traverse. It is used to establish third order and fourth order points.5km of traverse leg is taken into account.

3 Tertiary traverse: It is conducted under the controlled area of the primary and secondary order traverse. It is used for cad-astral purposes.

4 Fourth order traverse: it is conducted under area of second order and third order traverse. It is used for engineering project.

2.3 Methodology

2.3.1 Instrument Used

- **Theodolite:** A theodolite is a precision instrument for measuring angles in the horizontal and vertical planes. It is used for finding differences in elevations and setting out works of roads, railways, etc.
- **Ranging Rod:** Ranging Rods are the rods of circular cross-section having the length of 5 m. They are used to range some intermediate points in the survey line.
- **Measuring Tape:** Tape is used to measure the height of instrument during our project.
- **Tripod stand:** A surveyor's tripod is a device used to support any one of a number of surveying instruments such as theodolites, total stations, and level.

Figure 8: Theodolite

Figure 9: Tripod stand

Figure 10: Meter

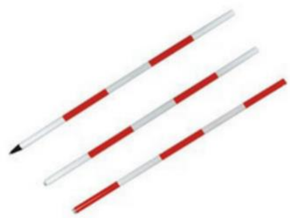


Figure 11: Ranging Rod

2.3.2 Study Area

The study area for traverse was carried out around Jiri Airport. A number of 12 traverse stations were established in this geographically diversified area.

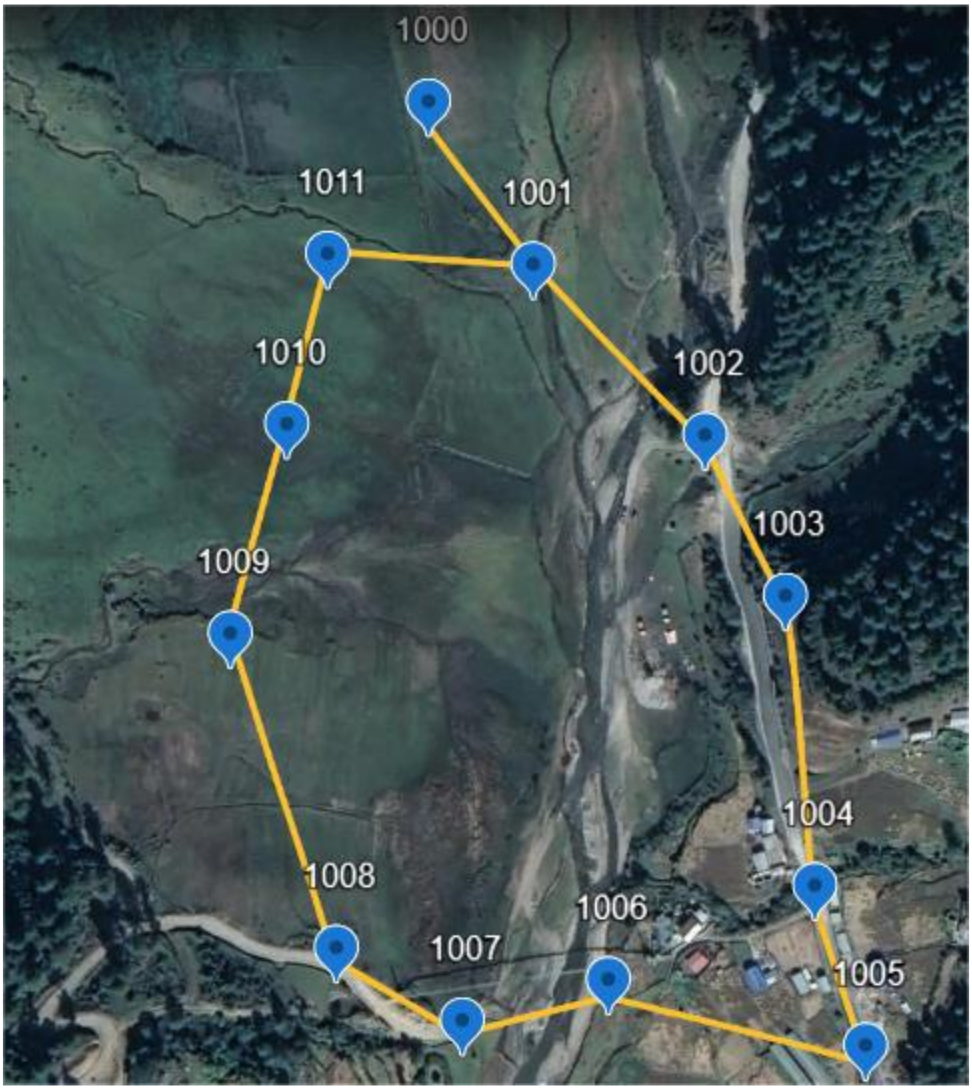


Figure 12: Study Area for Traverse

2.3.3 Specification

Table 2: Specification for Traverse

Particulars	**Specification**
Types of traverses	Closed traverse
Order of the work	Fourth order
Numbering	1001-1011
Time required for traverse	3 days
Name of instrument	ZIPP-02
Purpose	To establish Horizontal and Vertical control points
Unit of Measurement	Linear measurement: Meter Angular Measurement: Degree
Instrument used	Linear Measurement: Total Station Angular measurement: Theodolite
Linear Measurement Direction	Two-way, double measurement
Angular Measurement Direction	Clockwise
Number of sets for angular measurement	Three sets
Face to Face tolerance	1 minutes 42 seconds
Set to Set tolerance	20 seconds
RO to RO tolerance	!0 seconds
Angular Accuracy	75√n (n is number of traverse station)
Method of Angular Misclosure adjustment	Equal shift
Method of Linear Misclosure adjustment	Bowditch rule
Method of vertical Misclosure adjustment	Equal shift

2.3.4 Study Method/Design/Workflow

Figure 13: Methodological Flow Diagram

- **Reconnaissance**

  Reconnaissance is fundamentally the first and foremost field operation and is taken as one of the important phases of surveying. In this phase, we fully inspected the field to be surveyed with. We marked the position of 12 stations on the ground considering the visibility and commanding point of these points and traverse legs. Similarly, minor bushes and branches were cut down for inter-visibility of the marked stations.

- **Monumentation**

  We marked the selected stations on the ground with the wooden pegs around the permanent witness structure so that they could be recovered at any time during the project period. Altogether 12 control points were monumented.

1. **Signaling**

   Not all the monuments we've marked on the stations were visible from adjacent stations. For solving this deficiency, we used ranging rods erected vertically above the center of the stations as a signal during traverse.

- **Observation**

  Altogether, 12 stations were established in the project site initiating from 1000 and ending at 1011.

  In each station, three different types of observation were made in theodolite traversing namely,

i. Horizontal Angle

ii. Vertical Angle

iii. Height of instrument

For the angular measurement, following procedures were followed:

1. Setting up a tripod at a convenient height for the observer.
2. Setting up the theodolite followed by centering with optical plummet.
3. Leveling up initially with a circular bubble followed by leveling with plate level.
4. Cross Hair Focusing.
5. Image Focusing.
6. Sighting towards the target.

i) **Horizontal Angle**

- We carried out a tertiary traverse where three sets of horizontal angle reading were taken for each station. For the first set, the RO was set to 0°10'00". Similarly, for the second and third set of readings, the RO was set to 60°10'00" and 120°10'00" respectively. Three different sets of reading were taken to eliminate graduation error. The permissible misclosure for set-set reading was 60ccg.
- And, in each set, two face readings were taken; left face reading and right face reading. This was done by swinging the theodolite clockwise followed by transiting, to change the orientation of the vertical circle in theodolite. Two faces reading was taken to eliminate the collimation error from theodolite. The permissible misclosure for face-face reading was 1minutes 42 seconds.

ii) **Vertical Angle**

- In the field, we had only taken a single set of readings and that set includes both left face and right face readings. Only one set observation was taken in any set while observing horizontal angle. As like in horizontal angle measurement, to eliminate index error, both faces left and face right observation were taken. The index error was then distributed equally to both face left and face right reading to get corrected zenithal angles. Since we observed the vertical angle to determine the reduced level as a procedure of trigonometric leveling, only a single set of readings was taken as it is not one of the finest methods of determining reduced level. Vertical Angle is measured to determine the RL of traverse stations.

iii) **Distance**

- For the distance measurement, Total station was carried out and used under the supervision of faculties. Furthermore, we followed the anticlockwise path for taking observations in the closed loop of traverse stations.
- Thus, by computing these polar readings observed, the (x, y, z) coordinates were established.

- **Computation**

i) Computation of (x, y) coordinates

In each station, the average of three horizontal angles obtained from three sets of readings were calculated, which is the horizontal angle of those particular stations.

Having 11 stations in a closed traverse, the sum of traverse angles must have been

(2n+4) *right angle = ((2*11+4) *90°

=2340 where, n is the no. of stations

However, due to existence of some systematic errors, natural errors and random errors, the sum of exterior angles was found out to be 2340°01’02”, with angular misclosure 0°01’02” which is within the permissible misclosure 0°01’20” given by

$C= k\sqrt{n} =75\sqrt{11}$

Where, n is the number of angles and k is a fraction of the least count of transit theodolites. The misclosure was adjusted by the technique of Average Adjustment in each station angle. Since, we were provided with the bearing of traverse leg 1000-1001 i.e. 155°07'46 ", we easily calculated the departure (ΔE) and latitude (ΔN) of each traverse legs. However, the angular error obtained in the latitude and departure was corrected by applying the Bowditch Method in each leg.

For instance, the (ΔE, ΔN) (before adjustment) of leg 1000-1001 was (77.822, -67.2107).

But we'd the angular error in departure and latitude, 0.0343 and -0.0738 respectively.

And the perimeter of the traverse was 1031.548m, having the length of leg 1000-1001, 102.828m.

According to Bowditch adjustment method, we proceed the adjustment as follows:

Adjustment in Latitude of 1000= latitude misclosure*(length of 1000-1001) / (Perimeter of traverse)

Adjustment in Departure of 1000= (departure misclosure* Length of 1000-1001)/perimeter of traverse

Therefore, the adjusted ΔE is 77.818 with ΔN is -67.203 and remaining values were adjusted similarly in succession. Thus, we obtained corrected values of latitude and departure of each leg. Since, we were also provided with the coordinates of station 1000 and 1001, we derived the coordinates of station 1002 as:

E of 1002= E of 1001+ΔE of 1000-1001

N of 1002= N of 1001+ΔN of 1000-1001

And subsequently, we derived the coordinates of the remaining stations.

ii) Computation of RL (Elevation)

In each station, a single set of Zenithal Reading was taken. Both left and right face readings were taken at each of the adjacent targets (staff) from the instrument station and the zenithal angle observed was recorded. For the staff at a station, the sum of zenithal angle observed from the instrument station for left face and right face reading must have been equal to 360°. But due to the existence of some systematic errors, natural errors, and random errors, the sum in some cases fluctuated from 360°. But those within the permissible error were adjusted by Average Adjustment Method and were reduced to 360°.

2.3.5 Data Used

We've used the self-measured data collected in the field. However, the length of traverse legs, coordinates of station 1000 and bearing of leg 1000-1001 were provided by the faculties of the department.

Coordinate of station 1000 = Easting (424111.871) and Northing (3055836.523)

Bearing of line 1000-1001 = 155°07'46 "

Table 3: 3-D position of traverse stations (All data are in meters)

STATION	EASTING	NORTHING
1000	424111.871	3055836.523
1001	424168.048	3055715.337
1002	424245.867	3055648.132
1003	424290.275	3055570.773
1004	424297.378	3055456.59
1005	424319.337	3055377.177
1006	424198.698	3055411.154
1007	424128.834	3055399.721
1008	424074.812	3055432.202
1009	424043	3055543.052
1010	424088.794	3055598.300
1011	424069.892	3055692.008

2.4 RESULT

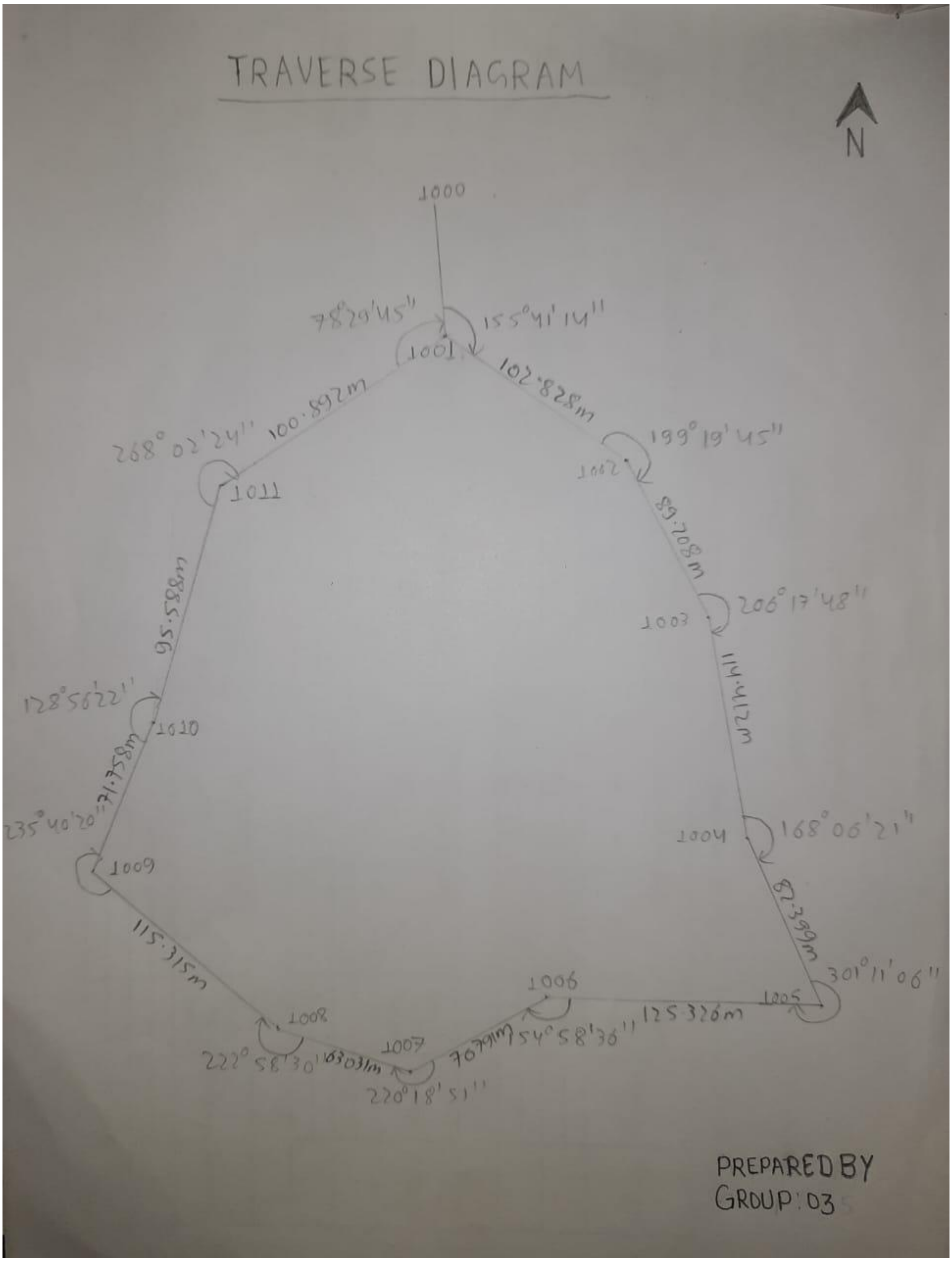
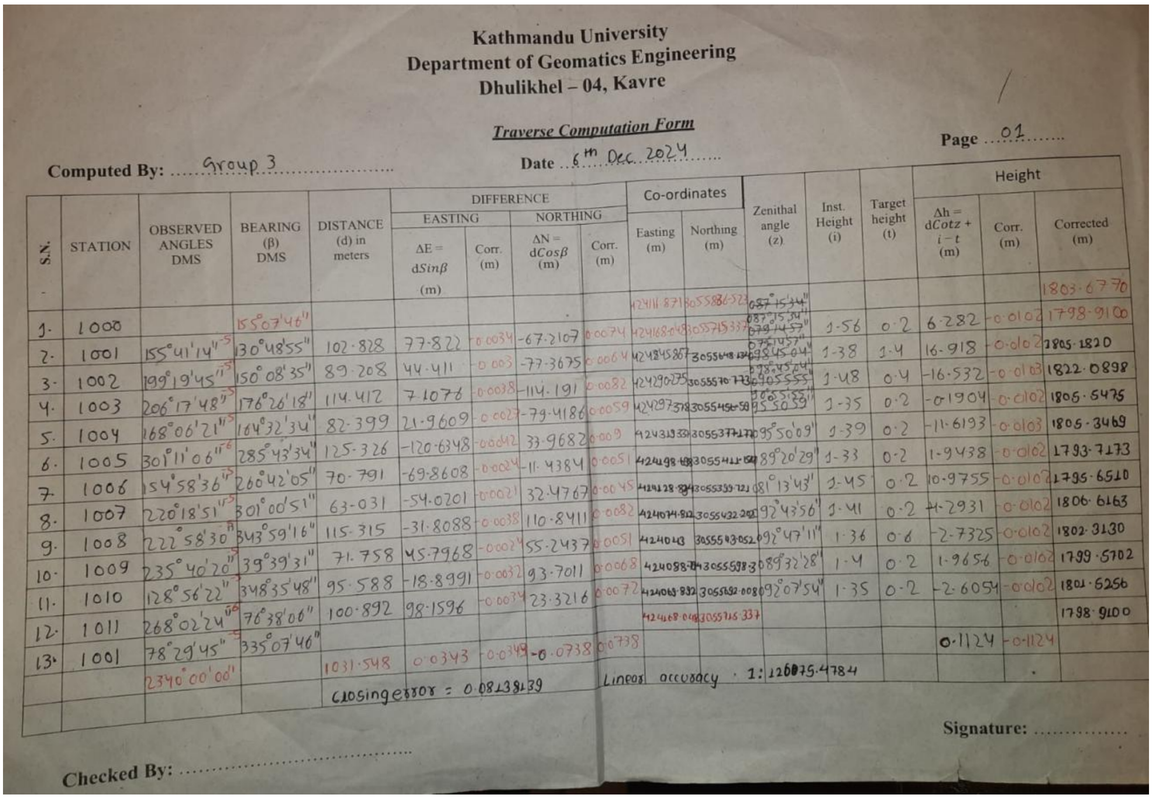


Figure 14: Traverse Loop (external angles)



Kathmandu University
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Traverse Computation Form

Computed By: Group 3 Date 6th Dec. 2024 Page 01

S.N.	STATION	OBSERVED ANGLES DMS	BEARING (β) DMS	DISTANCE (d) in meters	DIFFERENCE EASTING ΔE = dSinβ (m)	EASTING Corr. (m)	NORTHING ΔN = dCosβ (m)	NORTHING Corr. (m)	Co-ordinates Easting (m)	Co-ordinates Northing (m)	Zenithal angle (z)	Inst. Height (i)	Target height (t)	Height Δh = dCotz + i − t (m)	Height Corr. (m)	Height Corrected (m)
-																1803.6770
1.	1000		155°07'46"						424111.871	3055886.523	087°15'34"					
2.	1001	155°41'14" $^{-5}$	130°48'55"	102.828	77.822	-0.0034	-67.2107	0.0074	424168.048	3055715.337	087°15'34" 079°14'57"	1.56	0.2	6.282	-0.0102	1798.9100
3.	1002	199°19'45" $^{-5}$	150°08'35"	89.208	44.411	-0.003	-77.3675	0.0064	424845.867	3055648.134	079°14'57" 098°45'04"	1.38	1.4	16.918	-0.0102	1805.1820
4.	1003	206°17'48" $^{-5}$	176°26'18"	114.412	7.1076	-0.0038	-114.191	0.0082	424290.275	3055570.773	098°45'04" 090°55'55"	1.48	0.4	-16.532	-0.0103	1822.0898
5.	1004	168°06'21" $^{-5}$	164°32'34"	82.399	21.9609	-0.0027	-79.4186	0.0059	424297.378	3055456.58	090°55'55" 95°50'59"	1.35	0.2	-0.1904	-0.0102	1805.5475
6.	1005	301°11'06" $^{-6}$	285°43'34"	125.326	-120.6348	-0.0042	33.9682	0.009	424319.339	3055377.170	095°50'09"	1.39	0.2	-11.6193	-0.0103	1805.3469
7.	1006	154°58'36" $^{-5}$	260°42'05"	70.791	-69.8608	-0.0024	-11.4384	0.0051	424198.698	3055411.150	89°20'29"	1.33	0.2	1.9438	-0.0102	1793.7173
8.	1007	220°18'51" $^{-5}$	301°00'51"	63.031	-54.0201	-0.0021	32.4767	0.0045	424128.834	3055399.721	081°13'43"	1.45	0.2	10.9755	-0.0102	1795.6510
9.	1008	222°58'30" $^{-5}$	343°59'16"	115.315	-31.8088	-0.0038	110.8411	0.0082	424074.812	3055432.200	92°43'56"	1.41	0.2	-4.2931	-0.0102	1806.6163
10.	1009	235°40'20" $^{-5}$	39°39'31"	71.758	45.7968	-0.0024	55.2437	0.0051	424043	3055543.052	092°47'11"	1.36	0.6	-2.7325	-0.0102	1802.3130
11.	1010	128°56'22" $^{-5}$	348°35'48"	95.588	-18.8991	-0.0032	93.7011	0.0068	424088.774	3055598.3	089°32'28"	1.4	0.2	1.9656	-0.0102	1799.5702
12.	1011	268°02'24" $^{-6}$	76°38'06"	100.892	98.1596	-0.0034	23.3216	0.0072	424069.832	3055692.008	092°07'54"	1.35	0.2	-2.6054	-0.0102	1801.6256
13.	1001	78°29'45" $^{-5}$	335°07'46"						424168.048	3055715.337						1798.9100
		2340°00'00"		1031.548	0.0343	-0.0349	-0.0738	0.0738						0.1124	-0.1124	
				closing error = 0.08138139					Linear accuracy : 1:126075.4784							

Checked By: ....

Signature: ....

Figure 15: Traverse computation Form

2.5 Conclusion and Recommendation

Hence, the three dimensional position of 11 stations were obtained by theodolite traversing with the closing error of 0.08138139. The relative accuracy of our project was 1:12675.4784. Works of detailing, plotting the detail on tracing paper and later shifting it on permatrace was done with the supervision of coordinators.

2.6 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

3 CHAPTER THREE: TACHEOMETRY

3.1 INTRODUCTION

3.1.1 Background

Tacheometry is a rapid and economical surveying method by which the horizontal and vertical positions of point on the earth surface and the difference in elevation are determined using intercepts on a graduated scale and the directions are observed with a transit theodolite. The instrument which is used for Tacheometric purposes is known as Tacheometer. Tachometric measurements are of three types:

A. Stadia System: In this system, horizontal distance and difference of elevation are calculated from the staff intercept, horizontal and the vertical angle.

● Fixed hair method: In this method, vertical spacing between upper and lower stadia hair is kept fixed.

● Movable hair method: In this method, the vertical spacing between the stadia hair can be varied by moving the stadia hairs vertically.

B. Tangential System: In tangential method, horizontal distance and vertical distance from the instrument to the staff are computed from the observed vertical angle to the vanes fixed at a constant distance apart upon the staff.

C. Subtense bar system: It is a bar of accurate length mounted horizontally over a tripod with a levelling head. Subtense bar has targets at the two ends at a fixed distance. Horizontal angle is also measured between these two targets and stations under this system.

Among the above systems, we followed a fixed hair method under the stadia hair system for detailing.

3.1.2 Principle

The principle of tacheometry is based on the principle that the ratio of the perpendicular to the base is constant in similar isosceles triangles.

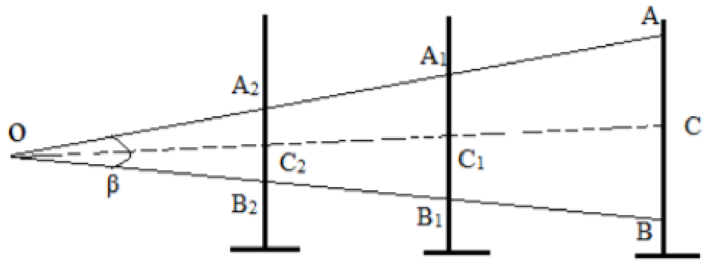


Figure 16: Principle of Tacheometry

In the figure, let two rays OA and OB be equally inclined to the central ray OC. Let A2B2, A1B1 and AB be the staff intercepts. Then,

OC2/A2B2 = OC1/A1B1 = OC/AB = constant k= $1/(2\cot\beta/2)$

3.1.3 OBJECTIVES

The primary objective of tacheometric surveying was:

- To prepare the topographic map of Jiri.

The secondary objectives to accomplish the primary objectives were:

- To save data of features inside the traverse network.
- Plotting the details and Contouring.

3.1.4 Study Area

The study area for traverse was carried out around Jiri Airport. Eleven stations, which were established during traverse, were used in tacheometry for the detail study of that area.

Figure 17: Study Area for Tacheometry

3.1.5 SCOPE OF WORK

Tacheometry is the detailed survey used to prepare the topographical map of the area. This project deals with the scope of map preparation to show the topography of the site. The prepared map will show elevation change using contour lines. Using the contour, our map can be used to find the area and volume. This map also includes the features of the site such as buildings, roads, electric poles etc. The map prepared is mono- colored and use of symbols is done to represent various features. Preparation of a multi-colored map is not under the scope of our project. Our prepared topographical map can be used as the base map of area for decision making for other engineering projects.

3.2 Literature Review

3.2.1 Technical Terms:

- **Horizontal Angle:** Angle made by the detail with the reference traverse line.
- **Horizontal Distance:** Distance between the Instrument station and the detail.

- **Tacheometer station:** It is the desirable station which commands a clear view of the area to be surveyed within the range of observations.
- **Anallatic lens:** It is an additional convex lens provided between the eye-piece and the object glass at a fixed distance from the object glass.
- **Contour interval:** The vertical distance between any two consecutive contours is known as contour interval. It is kept constant in the map.
- **Reduced Level:** Elevation of the detail with reference to the datum.
- **Line of sight:** It is an imaginary line passing through the optical center of the object glass and the intersection of the cross-hairs at the diaphragm.
- **Tachometer:** It is the theodolite which has stadia hair in the diaphragm**.**

3.2.2 Specifications:

Table 4: Specification for Tacheometry

Particulars	Specification
Order of the work	Fourth order
Purpose	Collect data for topographical map
Time	4-5 days
Maximum staff reading	2.7 meter
Minimum staff reading	0.3 meter
Additive constant	0
Multiplicative constant	100
Name of theodolite	ZIPP-02
Index contour interval	5 meters
Intermediate contour interval	1 meter

3.3 Methodology

3.3.1 Instrument Used

- Ranging Rod
- Tape
- Theodolite
- Staff
- Tripod stand

3.3.2 Instrument Check

When we get the instruments we have to take a random angle by face left and face right observation. The result obtained was:

- For the zenithal angle, the sum of left face reading and right face reading was 360 Degrees.
- For horizontal angle, left face reading and right face reading were differing by 180 Degrees.
- Additive constant and Multiplicative constants were checked and found to be 0 and 100 respectively.

3.3.3 Reconnaissance and Monumentation

The Project Area was visited and the feasibility of the area was studied. 7 control Points were selected considering the inter-visibility and length of traverse legs. At the time of reconnaissance, the whole area to be detailed was investigated and the features to be taken in concern were detected and a rough sketch was drawn. The places to throw the offsets were also estimated. The control points were monumented with wooden pegs.

3.3.4 Observation

Observation was done by Tacheometry method. The Staff intercept, Horizontal angle and Zenithal angle of the details were observed with the Theodolite and Height of Instrument was measured with the measuring tape. Some offsets were taken for the details that were not visible from the control points. Similar readings were taken from those offsets. We've taken 4 offsets in total to cover the whole area inside the traverse loop.

3.3.5 Computation

When the horizontal and zenithal angles and staff intercept of the details were observed, their horizontal distance and R.L were computed using the formulae:

Horizontal distance = $K * (t-b) * \sin^2 Z + C * \sin Z$

Reduced Level (R.L.) = R.L. of station + ΔH + H.I. – m

Where,

$\Delta H = K*(t-b)*SinZ*CosZ + C*SinZ$

t= top reading of stadia hair

b= bottom reading of stadia hair

m= middle reading of stadia hair

H.I. = Height of Instrument

Z= Zenithal angle

C= additive constant

K= multiplicative constant of theodolite

3.3.6 Plotting and Map Preparation

After computation of horizontal distance and RL of the details, the plotting of data was started. In this phase the data were firstly plotted on permatrace using pencil. At first the paper scale was calculated as 1:500 and origin was set. Map was traced on the permatrace. The points and the features of the tracing paper were pricked on the working face of the permatrace paper. Then the traverse station was plotted. By taking the reference of the traverse station other details were plotted in permatrace using the same scale. At first physical features were drawn and index contours having interval 5m and intermediate contours having contour interval 1 m is plotted. Legends were provided to indicate the specific features like buildings, roads, boundary walls and others. Title of the map was provided on the top of the paper and the authors on the bottom of the permatrace paper.

3.4 Results

From the given Project, following results were obtained:

Detailing: Horizontal and zenithal angles and staff reading of the details were measured.

Computations: The horizontal distance and R.L. (Reduced Level) of the details were computed with respect to the coordinates of the station by various formulae.

Plotting: The details were plotted in polar form (i.e. in distance and angle) in the Permatrace paper in the scale of 1:500. Protractor printed in paper was used for angular measurement.

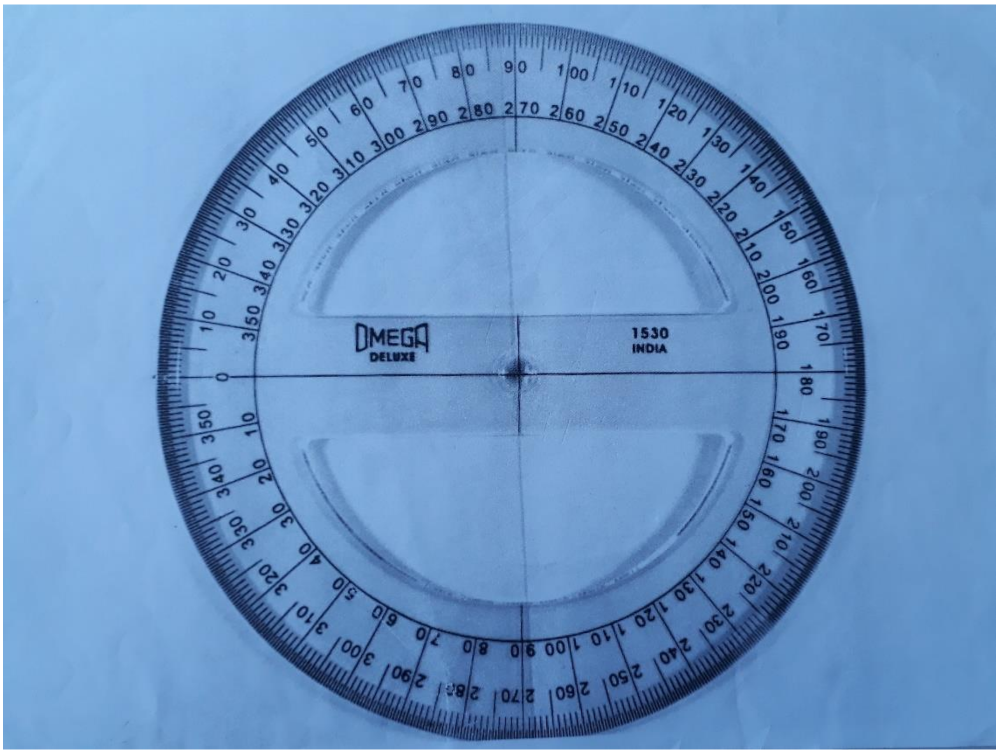


Figure 18: Protractor

Contouring: Contour lines with contour intervals of 1m were made in the map with the help of spot height by interpolating the contours.

Drawing Legends and Symbol: Different details were represented by drawing symbols and legends.

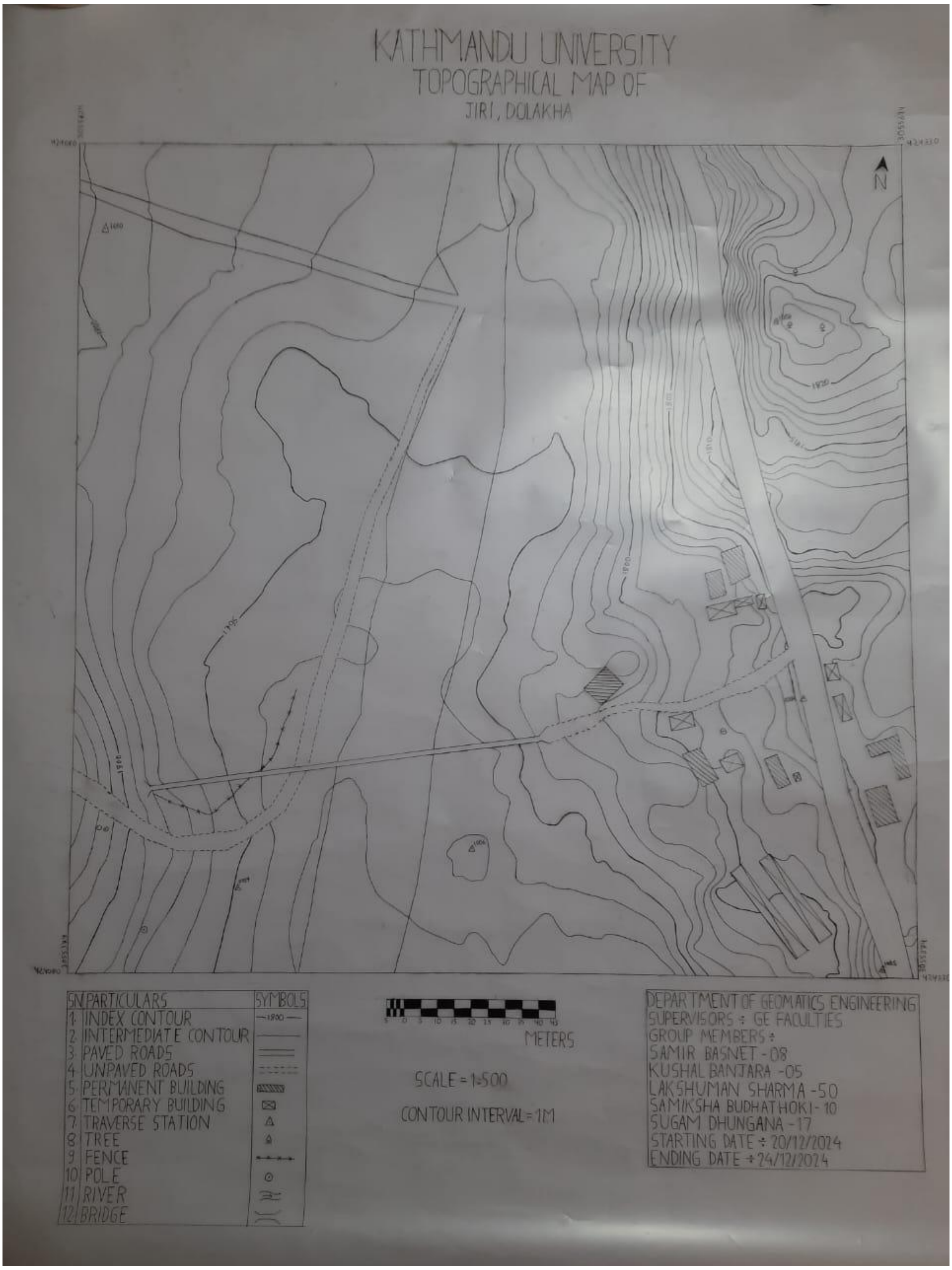


Figure 19: Final Topographical Map

3.5 Conclusion

For the preparation of a topographic map, detailing of the site is essential. For that purpose, a method of Tacheometry was adopted. It is a very rapid method for detailing. For tacheometry survey, coordinates of 11 traverse stations were plotted and then the details observed were plotted. Overall, the project was performed finely and the result plotted was also informative. We have completed all field work and required computations under supervision of supervisors. Finally, the topographical map of Jiri Airport was prepared on a 1:500 scale.

3.6 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

4 CHAPTER FOUR: TRIANGULATION

4.1 INTRODUCTION

4.1.1 Background

The method of measuring the angle of a chain of triangles produced by stations marked on the earth's surface is known as triangulation.

Principle of triangulation

- If all three angle and the length of one side of a triangle are known then by simple trigonometry the length of remaining sides of the triangle can be calculated.
- Again, if the coordinate of any vertex of the triangle and azimuth is known then remaining vertices may be computed.

Purpose of triangulation

- Establishing accurately located control points for plane and geodetic surveys of large areas.
- Establishing accurately located control points in connection with aerial surveying.
- Accurate location of engineering such as Centre lines, terminal points and shafts for long tunnels and Centre lines and abutments for long span bridges.
- To establish horizontal control points for topographic survey.

4.1.2 Objectives

The main objective of project is:

- To establish fourth order horizontal control points by the method of triangulation.

Sub objectives are:

- To learn basics of triangulation.
- To measure horizontal and zenithal angles.
- To establish vertical control points by trigonometric levelling.

4.1.3 Scope and limitation

The process of creating a horizontal control point is called triangulation. By using the Triangulation method, we were able to establish six control points, which can then be utilized to densify the control points. Additionally, it can be applied to a variety of surveys, including engineering, topographical, cadastral, and many more. It can serve as a framework for details as well. It can be applied to determine the area needed for engineering projects.

4.1.4 Technical terms

a. **Main stations:** They are the stations that are used to carry forward the network of the triangulation.
b. **Well- conditioned triangles:** A triangle is called well-conditioned if its shape is such that any error in the measurement of an angle has a minimum effect upon the computed lengths. The triangles used in triangulation system should be well-conditioned. The best shape of the triangles is an isosceles triangle whose base angles are 54°14'.
c. **Subsidiary stations:** They are the triangulation stations that are used only to provide additional rays to intersect points.
d. **Pivot stations:** They are the stations at which no observations are made but the angles at them are used for the continuity of the triangulation series.
e. **Satellite Stations:** They are the stations which are selected close to main stations to avoid the intervening obstruction.
f. **Strength of figure:** The accuracy of a triangulation system depends not only on the methods and precision used in making observations but also on shapes of figures in the network. The accuracy of the shapes is measured in terms of the strength of figure. It plays a vital role while deciding layout in triangulation system to achieve a desired degree of accuracy. It can be calculated as:

$$L^2 = \frac{4}{3} d^2 \quad R$$

where d is the probable error of an observed direction in seconds of arc, and R is a term which represents the shape of figure. It is given by:

$$R = \frac{D-C}{D} \sum \delta_A^2 + \delta_A \delta_B + \delta_B^2$$

Where;

D = the number of directions observed excluding the known side of the figure,

δ_A, δ_B= the difference per second in the sixth place of logarithm of the sine of the distance angles A and B respectively. (Distance angle is the angle in a triangle opposite to a side), and C = the number of geometric conditions for side and angle to be satisfied in each figure. It is given by $C = (n' - S' + 1) + (n - 2S + 3)$

Where;

n = the total number of lines including the known side in a figure,

n' = the number of lines observed in both directions including the known side,

S = the total number of stations, and

S' = the number of stations occupied

In any triangulation system more than one route is possible for various stations. The strength of figure decided by the factor R alone determines the most appropriate route to adopt the best shaped triangulation net route. If the computed value of R is less, the strength of figure is more and vice versa.

4.2 Methods

4.2.1 Instruments Used

- Theodolite with tripod
- Wooden pegs
- Ranging rods

4.2.2 Area of project

The project area is located at Dolakha district. It covers various important hills of Jiri area.

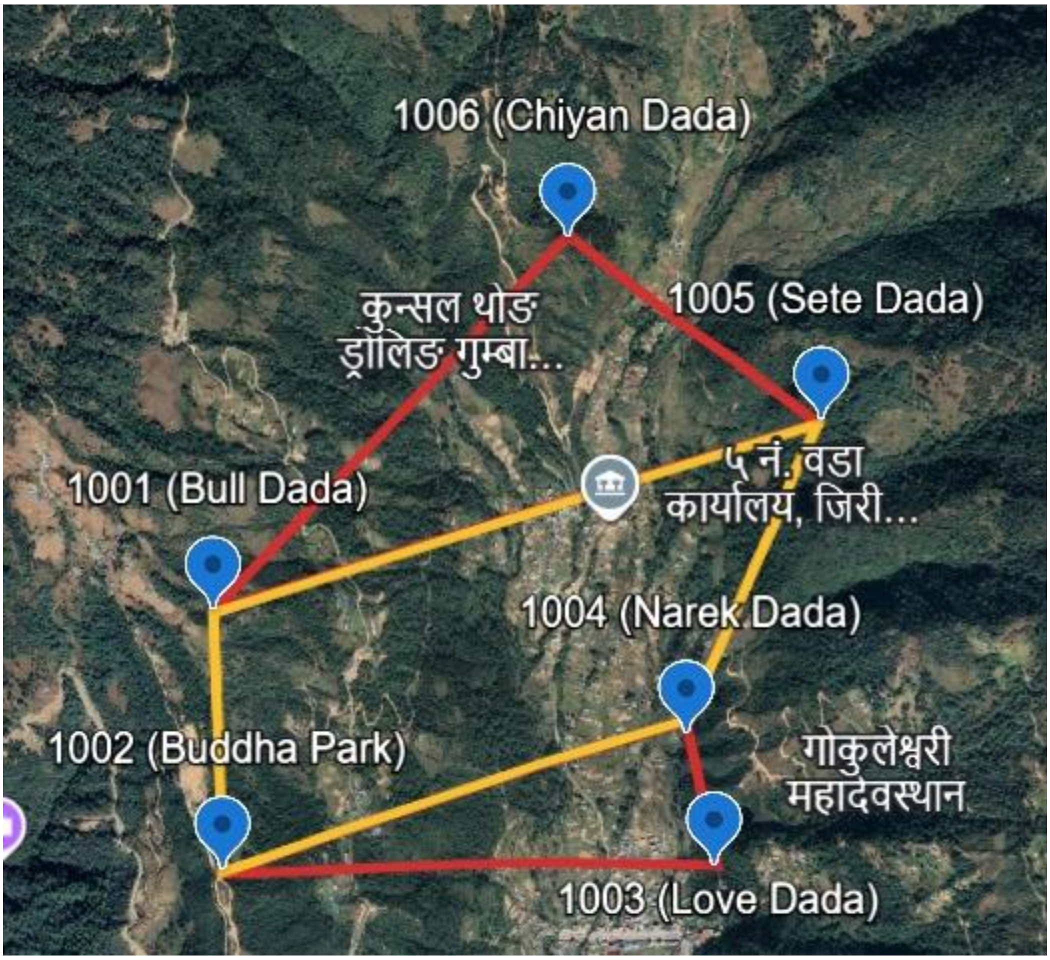


Figure 20: Study area for triangulation

4.2.3 Specification

Order of work: Fourth order

Type of triangulation: Braced quadrilateral with simple triangle

Numbering: fourth order numbering (1000-1005)

Angular unit: Degree, minute, second

Linear unit: meter

No. of sets: 3(000,060,120)

Least count: 1” (angle)

Least count: 1mm (distance)

Angular observation: Theodolite (ZIPP-02)

Distance observation (Base line): Total station

4.2.4 Methodology

- Reconnaissance and planning

  Planning was done with the survey's goal, available instrument resources, and human resources in mind. The entire region was walked over in order to inspect the area that needed to be surveyed. The inter-visibility between stations was examined during the reconnaissance. Details regarding the route for convenient access to the instrument stations were given. The stations were selected to create a well-maintained triangle.

- Monumentation and D-card

  All the control points selected on the reconnaissance phase were monument with the help of marked wooden pegs. Information like station name and number, type of monument and its diagram, field sketch and description were filled to reach the station.

- Angular measurement

  I. **Horizontal angle measurement**: Three sets observation were taken in 000˚10'00", 060˚10'00", 120˚10'00".RO was taken on previous station and interior angle was measured between RO and next stations, finally RO was closed. To eliminate collimation error, both face left and face right observation were taken.

  II. **Zenithal angle measurement**: Only one set observation was taken in any set while observing horizontal angle. As like in horizontal angle measurement, to eliminate index error, both faces left and face right observation were taken. The index error was then distributed equally to both face left and face right reading to get corrected zenithal angles.

  III. **Linear measurement**: EDM (Electromagnetic distance measurement) instrument was used for measuring distance of base line.

4.2.5 Computation

Arithmetic check:

a. For each set calculation the sum of angle in face left and face right position. Divide the sum by 2 and deduce full circle if necessary.

b. Calculate sum of RO's of each set and multiply by number of directions. Deduce full circle if necessary.

c. Calculate the sum of all reduced angle and multiply it by number of sets. Deduce full circle if necessary.

d. $\sum a = \sum(b + c)$

Adjustment of observed angle:

- Angle condition

a. Sum of three angles of plane triangles should be equal to 180˚.

b. Sum of the eight angles of braced quadrilateral should be equal to 360˚.

c. $Ø1 + Ø2 = Ø5 + Ø6$

d. $Ø3 + Ø4 = Ø7 + Ø8$

e. $\sum \log \sin Ø l = \sum \log \sin Ø r$

In above condition Ø1, Ø2, Ø3……… Ø8 are the angle of braced quadrilateral.

- Adjustment of braced quadrilateral

a) Determining the sum of the angle and subtract it from 360˚ to get total error. Distribute the discrepancy equally in all angle.

b) Determining the sum of angle ($Ø1 + Ø2 = Ø5 + Ø6$) / $Ø3 + Ø4 = Ø7 + Ø8$
If not equal, determine the discrepancy. Correct each angle by one –fourth of discrepancy.

c) Finding the value of log sin Ø l and log sin Ø r.

d) Determining the discrepancy m i.e. $m = \sum \log \sin Ø r - \sum \log \sin Ø l$

4.3 Output and analysis

Table 5: Output of Triangulation

STATION	EASTING	NORTHING	REDUCED LEVEL
1000	422928.985	3057187.220	2386.586
1001	422836.102	3058112.959	2412.803
1002	423738.490	3059006.448	2245.405
1003	424302.155	3058798.685	2204.125
1004	424275.262	3057998.114	2098.104
1005	424379.649	3057854.185	2109.294

4.4 CONCLUSION

Hence triangulation station of fourth order was established which can be further used for map making purposes i.e. cadastral mapping and topographical mapping. It can also be used for different engineering projects.

4.5 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling*.Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying*.Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

5 CHAPTER FIVE: INTERSECTION

5.1 INTRODUCTION

5.1.1 Background

The technique of identifying control locations without occupying them is called intersection. The position of inaccessible stations, like towers, temples, etc., can be ascertained by monitoring the direction towards the new stations from trigonometrical stations, whose coordinates are already known. However, this method can be applied to any fourth order station.

The coordinates of a second point can be calculated if the coordinates of the first point are known, along with the azimuth and distance to the second point. By applying the bearings α and β to P from known points A and B, whose coordinates are EA, NA, and EB, NB, one can determine the coordinates of P.

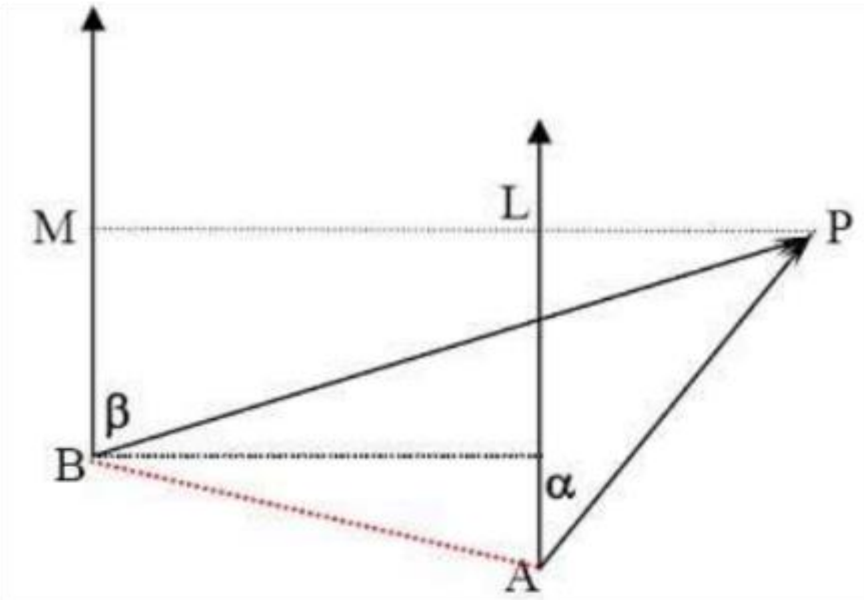


Figure 21: Intersection

In the above figure, A and B are known point with coordinates (EA, NA) and (EB, NB) respectively and that of P be (EP, NP) coordinate of unknown point. we determine the coordinate of P by sighting point, keeping instruments at A and B. The coordinate of P is calculated using the equation

$$EP = \frac{NB - NA + EA \cdot \cot\alpha - EB \cot\beta}{\cot\alpha - \cot\beta}$$

$$NP = \frac{EB - EA + NA \cdot \operatorname{Tan}\alpha - NB \operatorname{Tan}\beta}{\operatorname{Tan}\alpha - \operatorname{Tan}\beta}$$

5.1.2 Objective:

The main Objective of the project was:

- To establish horizontal control points of fourth order by the method of intersection.

The sub objectives of the project are:

- To learn about the basis of intersection.
- To know the position of inaccessible points.

5.1.3 Specifications

Order of work: Fourth order

Numbering: fourth order numbering

Angular unit: Degree, minute, second

Linear unit: meter

No. of sets: 2 sets (000°,060°)

Least count: 1” (angle)

Angular observation: Theodolite (ZIPP02)

Face to face tolerance: 2 cg

Index error tolerance: 5 cg

Set to set tolerance: 60 ccg

RO to RO tolerance: 30 ccg

5.2 METHODOLOGY

5.2.1 Study area

The location chosen for the intersection was not far from Jiri Airport. Attached below is a map that shows the position of the intersection:

Figure 22: Study area of intersection

5.2.2 Instrument Used

- Ranging rod with tripod
- Theodolite with tripod
- Wooden pegs

5.2.3 Methodological Flowchart

Figure 23: Methodology Flowchart

- Reconnaissance and planning

During planning, available instrument resources, human resources, purpose of survey was kept on the mind. The area to be surveyed was examined by walking over the entire area. During the reconnaissance, the inter-visibility of unknown point from known stations was checked as well as inter-visibility among stations were checked.

- Observation

We measured the horizontal angles between the unknown location and the known stations while we were observing. Additionally, the angle taken at a known station is displayed in the table below:

Table 6: Observed angles

Instrument at	Angle	Observed
1000	A	44°48'08"
1001	B	31°15'07"

- Computation

The coordinates of an unknown point (P) were determined using the coordinates of two reference stations:

Table 7: coordinates of known point

Station	Easting (m)	Northing(m)
1000	424168.048	3055715.337
1001	424069.892	3055692.008

As the coordinate of A and B were known, so we can calculate the bearing of the line AB using the formula:

Bearing of line = $\tan^{-1}(\Delta E/\Delta N)$

5.3 RESULT AND DISCUSSION

After all this, the final coordinates of unknown point were calculated as P(424082.188,3055382.468).

5.4 CONCLUSION

The major objective of theodolite intersection was to calculate the coordinates of unknown point P which was achieved.

5.5 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

6 CHAPTER SIX: RESECTION

6.1 INTRODUCTION

6.1.1 Background

Resection is a method for determining the position of a point by observing the directions from that point to points of known position. It's a graphical or analytical process that involves drawing lines of position from known points to the point being determined.

"Fixing the position of an unknown station by sighting to three known stations" is the resection principle.

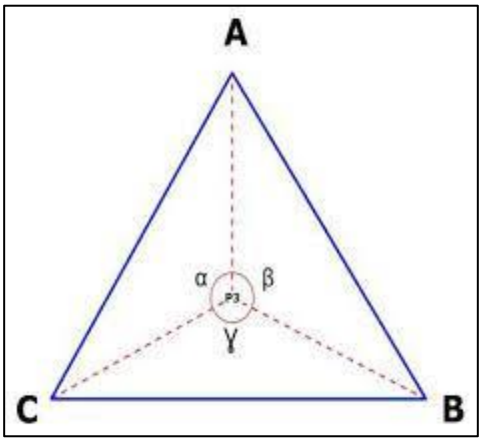


Figure 24: Resection

6.1.2 Objective

Resection's primary goal is to determine an unknown point's coordinate by utilizing an analytical method to sight three known points from the unknown position.

6.1.3 Method of Resection

Some methods of resection are below:

1. **Dr. T.L. Thomas's Method:** The coordinate of the unknown point P is calculated using this method as:

$$N_p = N_A + ZW/(V^2 + W^2)$$

$$E_p = E_A + ZW/(V^2 + W^2)$$

Where, V = ΔE1 Cot ∝ - ΔE2Cot (β + ∝) + (NC – NB)

W = ΔN1 Cot ∝ - ΔE2Cot (β + ∝) + (EB – EC)

Z = X Cot ∝ - X Cot (β + ∝) + Y Cot ∝ Cot (β + ∝)

Y = ΔE1ΔN2 - ΔN1ΔE2

X = ΔE1ΔE2 - ΔN1ΔN2

ΔE1 = EB – EA

ΔN1 = NB – NA

ΔE2 = EC- EA

ΔN1 = NB – NA

2. **ϕ - 45 method:**

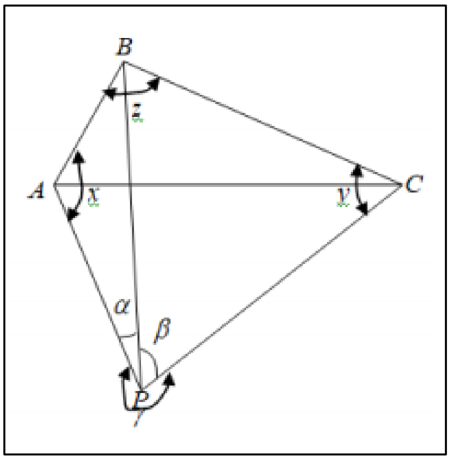


Figure 25: ϕ - 45 method

In the figure A, B and C are known stations and P is the unknown station. Angle APB (α), BPC (β) and angle ABC (z) can be computed as:

$$tan(y - x)/2 \ = \ tan(\phi \ - \ 45° \ ) \,.\, tan(y + x)/2$$

Where,

tanϕ = ($ABSin$β)/($BCSin$∝)

We compute the value of x and y and calculating the bearing of BP we calculate the coordinate of P.

3. **Bessel's Method:** The coordinate of unknown point I is calculated as:

$$E_I = \frac{E_B cot\alpha - E_C \cot\beta + N_B - N_C}{cot\alpha - cot\beta}$$

$$N_I = \frac{N_B tan\alpha - N_C \tan\beta - E_B + E_C}{tan\alpha - tan\beta}$$

Where, the coordinates of A, B and C are is (EA,), (EB,) and (EC,) respectively.

4. **Tienstra method:**

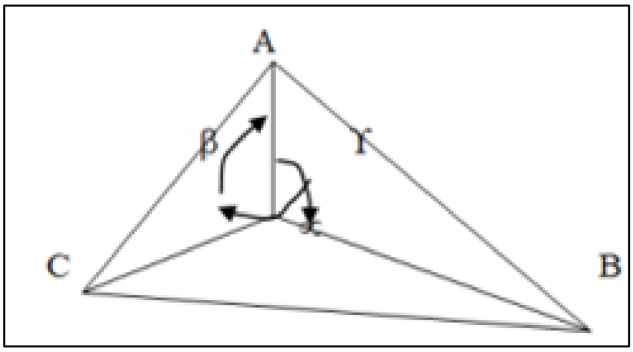


Figure 26: Tienstra or Barycentric method

A, B and C are the known points and P is the unknown point. We calculate the coordinate of P using formulas:

$$E_p = \frac{K_1 E_A + K_2 E_B + K_3 E_C}{K_1 + K_2 + K_3}$$

$$N_p = \frac{K_1 N_A + K_2 N_B + K_3 N_C}{K_1 + K_2 + K_3}$$

Where,

$$\frac{1}{K_1} = cotA - cot\alpha$$

$$\frac{1}{K_2} = cotB - cot\beta$$

$$\frac{1}{K_3} = cotC - cot\Upsilon$$

5. **Analytical Method**: This method was applied in the field to calculate the coordinate of unknown point.

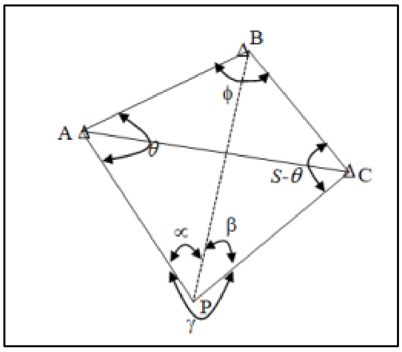


Figure 27: Analytical Method

In the figure A, B and C are the known stations and P is the unknown station. Angle ABC (ϕ) can be computed and APB (∝), angle BPC (β) can be measured. Similarly, side AP, BP and CP can also be measured and AB, AC, BC can be calculated.

Angle BAC (θ) can be calculated as:

$$Cot\theta = (Q + CosS)/SinS$$

Where, S = 360-ϕ - β - ∝

$$Q = (ABSin\beta)/(BCSin\propto)$$

After knowing the value of θ the bearing BP can be computed, and the coordinate of P can be calculated using the principle of traversing.

Note: This method fails in P lies on circumference of a circle passing through A, B and C.

6.1.4 Specifications for Resection

Table 8: Specification for resection

Particulars	Specifications
Angular unit	Degree, minute, second
Linear Unit	meter
No of sets	2 sets (000°,060°)
Least count	1” (angle)
Face to face tolerance	1min 42sec
Set to set tolerance	20 seconds
RO to RO tolerance	10 seconds

6.2 METHODOLOGY

6.2.1 Study Area

The site selected for theodolite resection was inside the premises of Jiri Airport. From there three stations were clearly visible. The map of that area is attached below:

Figure 28: Study area of Resection:

6.2.2 Instruments used

The instruments that were used during resection are mentioned below:

a. Digital theodolite with tripod.
b. Ranging rods.
c. Measuring tape.

6.2.3 Methodological Workflow

The chart showing the stepwise workflow of resection is shown below:

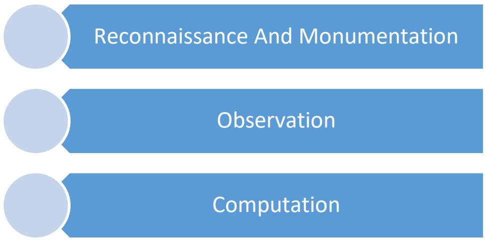


Figure 29: Methodological Workflow

I. **Reconnaissance and Monumentation:** As the known points for resection was taken among traverse stations so the area was already familiar to us. Thus, the recce was done quickly. The inter-visibility factor was kept in mind. Before starting the work, it was strictly checked that the point did not lie inside the perimeter of danger circle.

II. **Observations:** During observation, we took horizontal angles as well as zenithal angles. The analytical method of theodolite resection was used to calculate the coordinate of unknown point P. The table showing the known coordinates of station 2001, 2002 and 2003 is given below:

Table 9: Known coordinates

Station	Easting(m)	Northing(m)
1000	424111.871	3055836.503
1001	424168.048	3055715.337
1011	424069.897	3055692.008

And the observed angles are:

Angle α = 127°41'07"

Angle β = 103°56'5"

Angle γ =128°22'48"

6.3 RESULTS AND DISCUSSIONS

After the completion of calculations, the coordinate of unknown point was found to be P (424122.0432,3055830.166).

6.4 CONCLUSION

The purpose of theodolite resection was to calculate the coordinate of unknown point by sighting to three known points, which was successfully done.

6.5 REFERENCES

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

7 CHAPTER SEVEN: CHAIN SURVEYING

7.1 INTRODUCTION

7.1.1 Background

The land surveying technique known as chain surveying, or linear surveying, uses solely linear measurements. There are no angular measurements made. Chain surveying is utilized for small, open-space areas with a few basic characteristics. Measuring the lengths of the lines drawn out in the field is the most basic form of surveying. Only a chain, tape, and a few range rods are needed for chain surveying, and even a novice surveyor may operate them with ease. This surveying technique is rather straightforward and has a wide range of uses. However, huge areas with plenty of details are not a good fit for chain surveying.

A natural or man-made feature at or close to the ground surface is referred to as a detail in surveying. It includes

(1) hard details like walls, roads, and buildings

(2) soft details like trees, flora, and rivers

(3) subterranean details, such water mains and sewers

(4) above details, like telephone and power lines

Relies and undulations on the ground are not considered details. Chain surveying is used to prepare maps of the area to show different details or to secure data for precise description and boundary marking of a piece of land.

7.1.2 Objective

The fundamental purpose of chain survey is stated below:

a) To take the details of that area.

The secondary objectives of chain surveying is

a) To have basic understanding of the instruments used and chain surveying.

7.1.3 Types of chain

a. **Gunter's Chain**

Gunter's Chain is a chain that is 66 feet long and has 100 links, each of which is 0.66

feet long. This chain is helpful for measuring regions in arcs and length in miles. Ten squares of chain equal one arc in this case.

b. **Metric Chain**

This chain is made up of links, which are 4 mm diameter galvanized mild steel wire. After that, the ends of the links are bent into a loop and joined by three oval rings, which give the chain its flexibility. A tiny brass ring is distributed every one meter. Every five meters, it counts fixed. There are 100 links in this chain, each 20 cm long.

c. **Chain of Engineers**

Engineer's chain is a 100-foot chain containing 100 links, each of which is one foot long. These chains have brass tags attached at ten-link intervals, and their length is measured in feet and their area is measured in square yards.

d. **The revenue chain**

This chain has 16 links overall and is 33 feet long. For tiny areas, they are mostly utilized for precise measurements in feet and inches. In cadastral surveys, it is frequently used to measure fields.

7.1.4 Types of tape

A. **Linen tape or cloth:** This kind is composed of synthetic materials or woven linen. It can reach lengths of 10-30 m and widths of 12-15 mm. Such tape's primary drawback is its susceptibility to moisture damage and shrinkage.

B. **Metal tape:** To lessen stretching, copper or brass wires are braided longitudinally into linen tape. Their average length is between 20 and 30 meters. Offsets are measured with such tapes.

C. **Steel tape:** Typically, this kind of tape is 6 mm broad and 1–50 m long. At the end of this tape, which stands for zero, is a brass ring. They are not utilized on the ground since they are susceptible to rust.

D. **Invar Tape:** These tapes, which are composed of 65% steel and 35% nickel, are utilized for the most precise tasks. Compared to other tapes, this one is less impacted by temperature. They come in lengths of 30, 50, and 100 meters, with a width of 6 mm. It needs to be used with great care because it is soft and readily deforms.

7.1.5 Technical terms

I. **Main Survey Stations**: Main survey stations are conspicuous locations that are joined to form triangles by survey lines. These are the starting and finishing points of survey lines. The primary stations are A, B, C, and D in figure below.

II. **Verify the lines:** These are the lines that are used to verify that the triangles' frames are accurate. Plotting the triangle does not require the check line. When the region is being surveyed, a check line is measured in the field. The length of the check line is scaled out from the drawing and compared with the measured length in the field after the plan has been plotted using the survey lines. The length of the check line displayed on the plan should match its measured value if the survey lines were measured and drawn correctly. The check lines in Figure are lines CE and AF. Another name for check lines is proof lines.

III. **Lines of Survey:** Survey lines, also known as chain lines, connect the primary survey sites. The survey lines in Figure below are AB, BC, CD, DA, and BD.

IV. **Offsets:** The lateral distances measured from the survey lines are called offsets. The location of details is ascertained by taking offsets from the survey lines. There are two kinds of offsets.

1. Perpendicular offset

2. Oblique offset

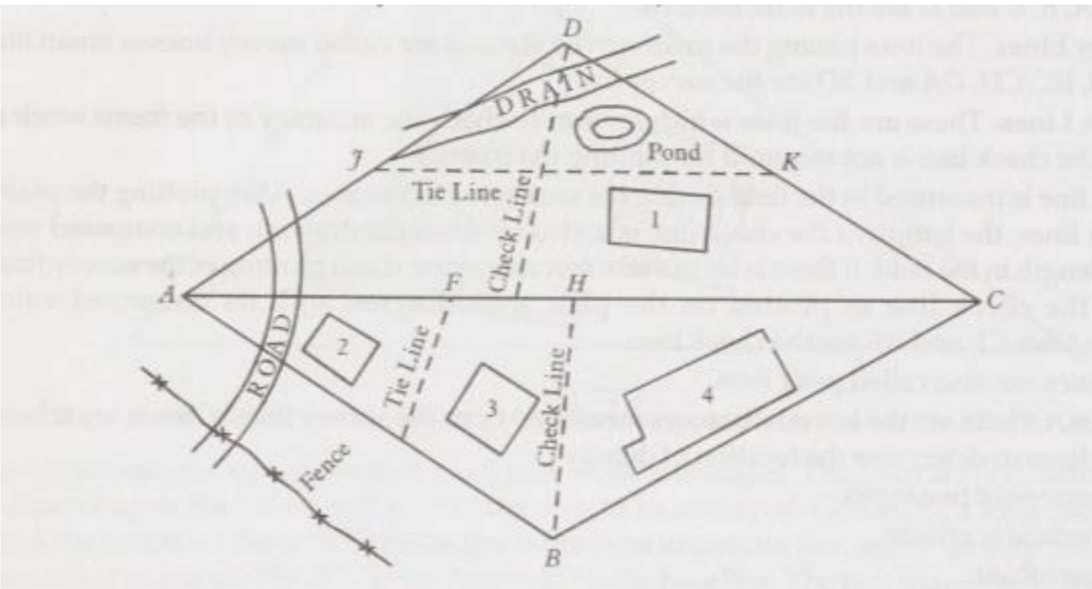
V. **Ties with range:** The oblique offsets made along a building's wall line are known as range ties, and Figure illustrates these as well.

VI. **Use tie lines**: It is necessary to take long offsets if the point of detail is quite far from the chain line. The lines used to find details in order to prevent lengthy offsets are known as tie lines.

VII. **subsidiary stations:** The tie lines or subsidiary lines are run between these stations, which are situated on the survey lines.
The JK line's subsidiary stations are shown in figures J and K.

VIII. **Baseline:** A lengthy survey line is drawn through the center of the region that has to be surveyed. The base line serves as the foundation for the triangles.

The basis line in the figure is line AC. Another name for the base line is the back-bone line.



7.1.6 Specifications

Table 10: Specifications for chain survey

Particulars	Specifications
Type of chain	Metric Chain
Number of Links	100 links
Distance per link	20 centimeters
Least count	20 centimeters

7.2 Methodology

7.2.1 Study area

All of the students were taken to the Jiri-5 municipality in the Dolakha area for the survey field tour. Farm Road served as the primary route for the chain survey.The chain survey's overall length was 50 meters.

Figure 30: Study area of chain survey

7.2.2 Instrument Used

- Metric chain
- Tape

7.2.3 Method

1. **Reconnaissance**: To gather as much information as possible, a small area was set aside for chain surveying. A road, an electric pole, and a few temporary and permanent homes were allotted to the area.

2. **Survey line running:** A survey line was run to measure the separation between main stations, and offsets were used to detect nearby details. The ranging out of the baseline was initially completed. Next, After unfolding, the chain was placed on the line. First, the distance between the main station and the offsets was measured, and then the ground feature detail was measured. Offsets were measured using tape.

7.3 Result and discussion

The planimetric map was created by gathering region details using a chain and tape.

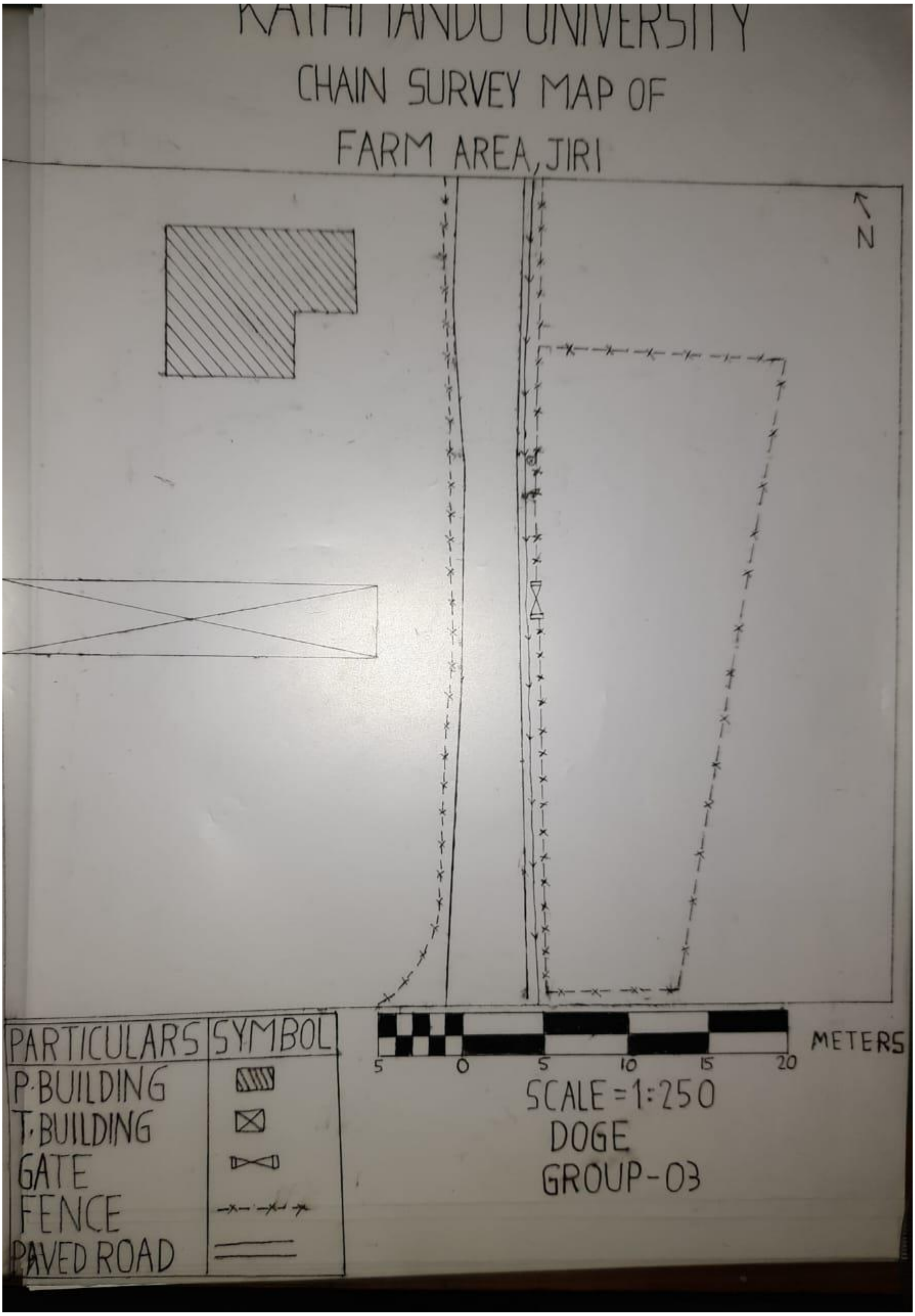
7.4 Conclusion

The area's details were gathered and a map was created using chain and tape.

7.5 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
- Duggal, S. K. (2004). Surveying. Tata McGraw-Hill Education.

7.6 Annex



Chain Surveying

8 CHAPTER EIGHT: COMPASS SURVEYING

8.1 Introduction

8.1.1 Background

Compass surveying is the area of surveying where survey lines' lengths and directions are established using a compass and directly chained or taped on the ground (Agor 1980). This area of surveying involves using a compass to identify the survey lines' orientations and measuring their lengths by taping or chaining them directly into the ground. It is utilized in expansive, undulating areas. One tool for determining the direction of lines is a compass. Because compass readings are so inaccurate, they are only utilized for simple, elementary surveying. Compass surveys are done by traversing. A traverse is made by connecting the plot points with a series of straight lines.

"Traversing, which involves a series of connected lines," is the foundation of the Compass Survey principle. A prismatic compass is used to measure the lines' magnetic bearing, and a chain is used to measure the area's length.

8.1.2 Objective

The primary goal of the Compass survey is:

a) To create a planimetric map by taking an area's details.

b) To gain knowledge about ranging.

c) To find our position on the map and get the traverse leg's bearing.

8.1.3 Comparison between surveyor and prismatic compass

Table 11: Comparison between surveyor and prismatic compass

Basis	Surveyor compass	Prismatic compass
Graduated ring	i) The Graduated ring is attached to the box and rotates along the line of sight. ii)The graduation has 0° at N and S and 90° at E and W	i) The graduated ring is attached with the needle and doesn’t rotate with line of sight. ii) The graduation has 0° at S, 90° at W, 180° at N and 270° at E.

Tripod	The instrument cannot be used without tripod.	The instrument can be held in hand also while making the observation.
Vanes	The eye vane consists of small vane with a small slit.	The eye vane consists of metal vane with a large slit.

8.1.4 Terms Used

1. **Meridian:** The reference line that survey lines are measured in relation to when calculating their horizontal angles.

2. **Bearing:** The measurement of the horizontal angle between the survey line and the reference meridian either in an anticlockwise or clockwise direction. True Meridian: The line where a plane intersects the earth's surface after passing through a specific location and the geographic (true) north and south poles is the true meridian running through that point.

3. **True bearing:** The line's true bearing is the horizontal angle, measured clockwise, between the line and the true meridian.

4. **Grid meridian:** When conducting a national survey, the actual meridian that passes through the center is occasionally used as the reference meridian for the entire nation. Grid meridian is the name given to such a meridian. The effect of meridians convergent is thus eliminated.

5. **Grid bearing**: Grid bearing is the horizontal angle formed by a line and the grid (central) meridian. The direction that the freely suspended magnetic needle indicates, independent of local attraction forces, is known as the magnetic meridians.

6. **Magnetic bearing:** A line's horizontal angle with the magnetic meridian is known as its magnetic bearing.

7. **Arbitrary meridian**: An arbitrary meridian is a meridian that is sometimes believed to be convenient while surveying a limited area. A line's horizontal angle with an arbitrary meridian is known as its "arbitrary bearing."

8. **Fore bearing:** The line's bearing as measured forward.

9. **Back bearing**: The line's bearing as measured from the rear.

10. **Whole circle bearing**: The horizontal angle measured from the north limb of the meridian in a clockwise orientation. It fluctuates between 0˚ and 360˚.

8.2 Methodology

8.2.1 Study area

Figure 31: Study area of Compass surveying

8.2.2 Instrument used

- Prismatic compass
- Chain
- Tape
- Tripod

8.2.3 Specification

- Angular unit: Degree
- Linear unit: meter
- Angular observation: Prismatic Compass

8.2.4 Methods

1. **Reconnaissance:** During reconnaissance the intervisibility between two stations was checked. The station was kept in fairly level ground. Every station was kept such that features or object on the ground are equally distributed.
2. **Measurement of length and bearing:** The distance between traverse legs were measured twice and mean of the length was taken as actual length.The compass was firstly fitted over the tripod and was leveled. The fore bearing and back bearing were measured with the help of surveyor compass. So, the result shown by the freely suspended magnetic needle was dependent upon the initial position of the graduated circle.
3. **Taking details:** For detailing of the features, ranging rod near the details was bisected by looking through the sight vanes. Angular measurements in whole circle bearings were observed and linear measurement from station to the details was measured with the help of tape.

8.2.5 Error adjustment

Adjustment of angle: All back bearing of traverse legs were converted to fore bearing and mean of these bearings were computed.

Adjustment of length: The error on length was adjusted by graphical method.

8.3 Results and discusssion

The compass survey was completed whose results are shown in the map of the farm area.

8.4 Conclusion

Finally, all the bearings were calculated, and the points were manually plotted onto the tracing paper and finally map was made on permatrace.

8.5 References

- Agor, R. (2012). Levelling. In *Surveying and Levelling.*Khanna.
- Punmia, D. B. (2011). Levelling. In *Surveying.*Laxmi.
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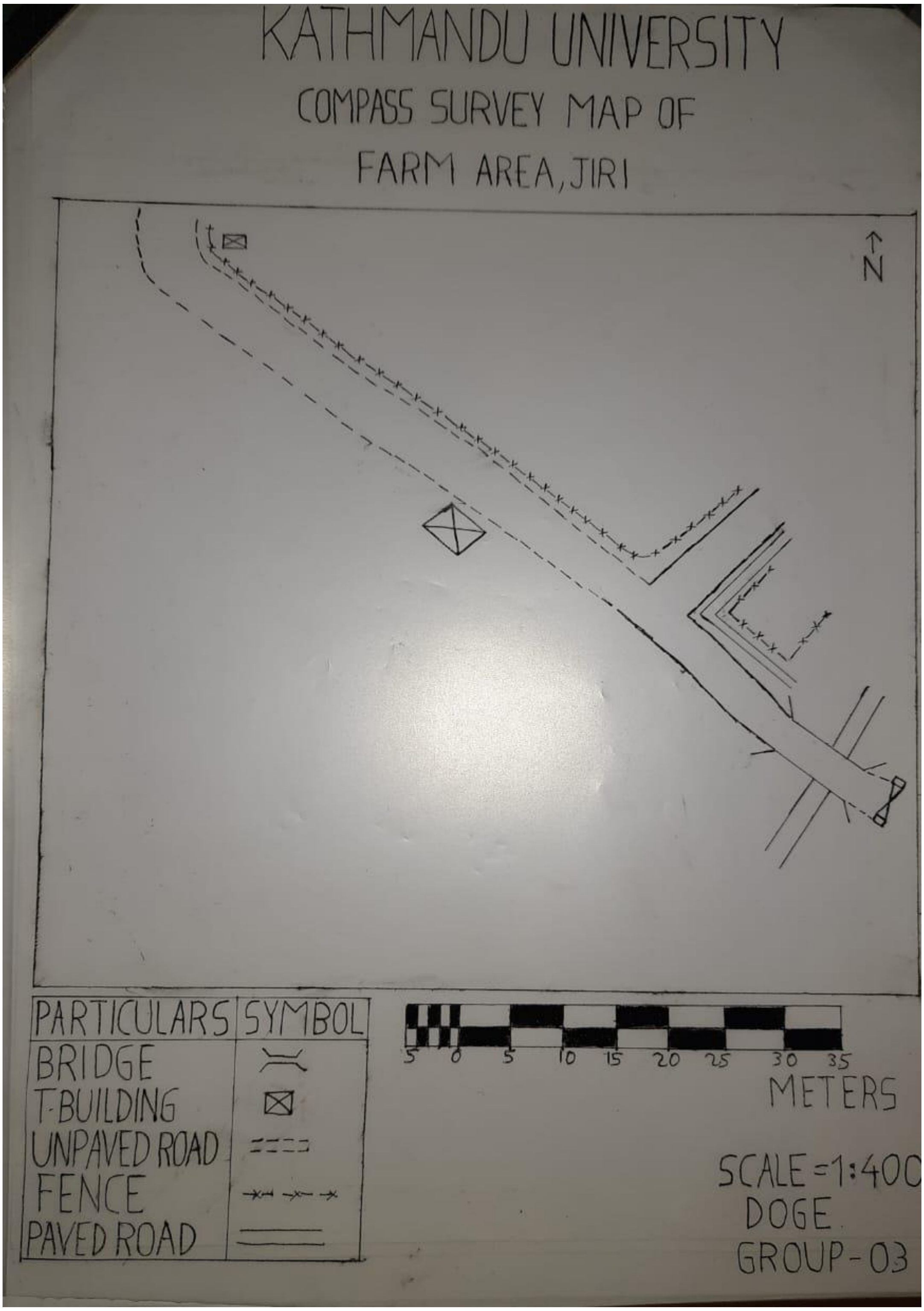
8.6 Annex



Compass Surveying